Recall, not verification, is the bottleneck when frontier LLMs elucidate molecular structures from real spectra

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Abstract

Given the molecular formula together with the infrared band list and ^{1}H/^{13}C shift lists exactly as reported in an open-access paper, how often does a frontier large language model (here, Claude) recover the correct molecular *constitution*? We find **28%** (top-1, n=194; 95% CI 22–35) — far below the near-100% implied by curated demonstrations.

The bottleneck is not the model's judgment but its *proposal*. Across the whole benchmark the model proposes the true structure for only **34%** of compounds; where it does, forward-verification — predicting each candidate's ^{13}C spectrum and re-ranking by agreement with the observed one — selects it **89%** of the time (58/65; 81% on the 37 where more than one candidate existed, §5.2). **Recall, not verification, is the wall.**

We contribute three things. **IRexp**, the largest openly redistributable collection of *experimental* infrared band lists (121,233 records, a third structure-linked), mined from open-access literature and released with its pipeline. **IRSpectra-Bench**, an open, blind, mechanically scored benchmark of 194 compounds on which accuracy splits sharply with structural complexity, and where a within-compound control shows reported numbers are sensitive to how a problem is posed. And a training-free **forward-verification** recipe, run over every benchmark compound. The two levers are distinct and we keep them apart: verification re-ranks a fixed candidate set and lifts top-1 from 28% to 30%, while *generating wider* is what moves recall — 31% to 41%, carrying top-1 from 23% to 30% on the arm where both were tested. Every top-1 step is directional rather than statistically resolved (§5.2, §5.3); the recall gain is the better-supported effect.

Those gains stop short of a ceiling, and the ceiling belongs to training-free elicitation rather than to the task: a small generator fine-tuned on IRexp lifts recall substantially and gives the highest full-benchmark accuracy we report (§5.6). But the wall moves rather than falls — the true structure stays outside the candidate pool for roughly half of compounds. Two independent contamination controls confirm the spectra do the work: masking them drops top-1 from 23% to 5% on the same compounds, and accuracy is flat in the publication year of the source paper (§4.6). Every result is single-vendor (Claude); a cross-vendor test is the key open question. The core protocol uses no model training and no paid API — two clearly-fenced probes (§5.4, §5.6) are the only exceptions — and all data, predictions, and code are released.

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1. Introduction

Determining a molecule's structure from its spectra is a central, time-consuming task in synthetic and analytical chemistry. The dominant machine-learning approach trains specialised encoder–decoder models to map spectra to structures; the recent *Spectro* model, for example, learns ^{1}H/^{13}C/IR → SELFIES from a corpus of 6,833 molecules.[1] In parallel, general-purpose LLMs have been reported to perform the same task off-the-shelf: a 2026 industrial white paper (a non-peer-reviewed company report) found that Claude Opus matched or beat commercial NMR-prediction software in the forward direction (structure→spectrum, ±0.08 ppm ^{1}H) and "recovered all eight simpler structures on every attempt" in the inverse direction (spectrum→structure).[2] We treat its numbers as a motivating claim to be tested against peer-reviewed benchmarks, not as an established baseline.

These results are striking, but the evaluation that supports them is narrow: 15 inverse problems on curated single-ring or two-fragment molecules, NMR only, with the seven harder targets additionally given the *starting-material structure* as a hint, and "recovery" scored leniently over three runs and three ranked candidates. The question a practising chemist actually faces — *take an arbitrary experimental spectrum from a paper and recover the structure* — is left open. Answering it requires (i) a large, diverse, real benchmark; (ii) a blind, reproducible scoring protocol; and (iii) honest accounting for the methodological choices that inflate or deflate the apparent score.

We provide all three. We build IRexp (§2), a literature-mined dataset that supplies the IR modality absent from prior LLM evaluations alongside paired NMR and resolved structures. We define a blind benchmark on it (§3) and measure inverse-elucidation performance under matched and mismatched methodologies (§4), isolating a large solver-methodology effect with a within-compound control. We then introduce forward-verification elucidation (§5), a training-free method that turns the model's *strong* direction (forward prediction) against its *weak* one (inverse regiochemistry), and use it to localise the bottleneck. The central finding is a sharp asymmetry: given a candidate set that contains it, the model *verifies* the correct structure 89% of the time, yet *proposes* it for only 34% of compounds — recall, not verification, is the wall (Fig. 1).

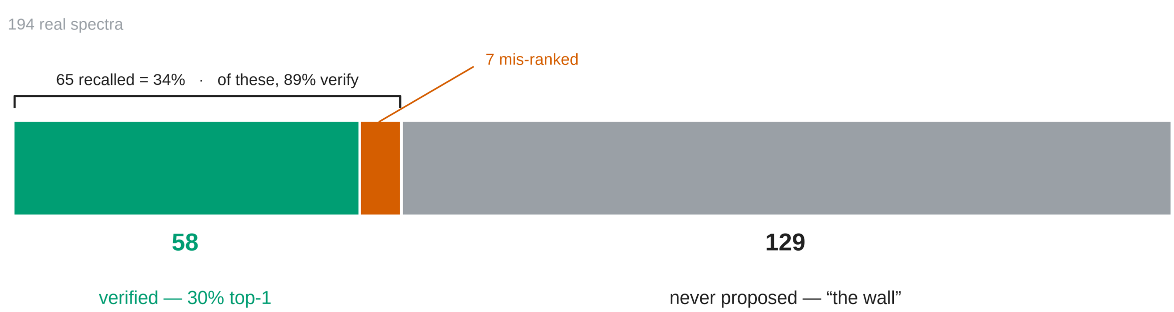

Fig. 1: The diagnosis in one glance: all 194 benchmark compounds as a single part-to-whole bar. The true structure is verified top-1 for 58 (green), recalled but mis-ranked for 7 (vermilion), and never proposed for 129 (grey) — *the wall*, 66% of the bar. Generation recall is 65/194 = 34%; conditional verification precision is 58/65 = 89%; their product is the 30% end-to-end top-1. The two rates have different denominators and are not differenced.

The end-to-end study design is summarised in Fig. S1. Throughout, the solver and verifier are LLM agents run under a consumer subscription — no fine-tuning, no API spend. That makes the protocol cheap to *re-run*, but cheap is not the same as reproducible: the subscription harness pins no model snapshot, exposes no temperature or seed, and carries an undisclosed product system prompt, so an outside group reproduces it distributionally rather than exactly (§8). What *is* exactly reproducible is the scoring — all predictions, ground truth and scorers are released, so every number in the training-free core (§3–§5.3, §4.6) regenerates from the released artifacts. The two trained probes are the exception: §5.4's HOSE lookup and GNN are built from an nmrshiftdb2 dump that we cannot redistribute, so those rows regenerate only once a reader supplies the same dump (see *Data and code availability*). Two trained probes (§5.6, §5.7) are reported separately as fenced complements. By construction this is an open-resource contribution in the remit of *Digital Discovery*: an openly licensed, structure-linked spectral dataset; a pre-registered, mechanically scored benchmark with released ground truth and scorer; and a training-free, zero-paid-API core pipeline (with two trained probes fenced in the SI) in which every main-text figure regenerates from the released code and predictions. We prioritise an honest, reusable measurement of where current models stand over a leaderboard-topping number.

Contributions.

1. **IRexp** — to our knowledge the largest openly *redistributable* collection of experimental IR **band lists** (121,233 records; 43,060 structure-linked; 33,201 IR+^{1}H+^{13}C+structure), released under permissive licences for bulk model development with a reproducible mining pipeline. The claim is scoped to that object type deliberately: view-only libraries such as SDBS hold more structure-linked *spectra*, and commercial libraries are larger still — neither is redistributable, and neither is what IRexp competes with (§2.1).
2. An **open multimodal benchmark** — to our knowledge the first blind, mechanically scored, complexity-stratified evaluation of frontier-LLM structure elucidation on real spectra — measured in depth on one model family (Claude), which reconciles the gap to optimistic prior reports. Crucially, the ground truth is experimental structures from the published literature, resolved deterministically (OPSIN/RDKit) and checked against the source articles mechanically (560/560 bands confirmed on a seed-fixed sample, §2.3; the *expert-chemist* review is prepared but not yet run, §7): no LLM curates the labels or scores the predictions. The model is the system under test, not the source of its own answers.
3. A **diagnostic decomposition** of LLM elucidation into recall and verification, showing that **recall — not verification — bounds current performance**, obtained via a training-free forward-verification probe run over **every** benchmark compound. The generate-and-forward-verify loop itself is prior art — NMR-Solver[3] implements it without an LLM and reports higher accuracy with a sharper predictor (§1.1); what is ours is the decomposition it enables, not the loop. It yields a bounded improvement (top-1 28%→30% on n=194; 23%→30% on the 60-compound arm where wide generation was also tested) but, by our own measurements, cannot exceed a recall/precision ceiling without sharper verification or 2D-NMR data.

1.1 Related work

Trained spectra→structure models. The dominant line trains sequence or graph decoders to emit structures from spectra: *Spectro* (^{1}H/^{13}C/IR→SELFIES)[1], the multitask CNN+transformer of routine 1D-NMR[4], and set/graph transformers such as NMRTrans[5]. These reach high accuracy *in-distribution* but require a labelled spectra→structure corpus, which is exactly the scarce resource our IRexp pipeline targets, and they are retrained per modality. Closest in spirit to our multimodal setting is **NMIRacle**[6], a generative model conditioned jointly on IR + ^{1}H + ^{13}C; it is a strong trained baseline, whereas our contribution is the *open experimental-IR data it (and others) can train on*, plus a *training-free* protocol and a blind benchmark to measure it.

LLMs as elucidators, and how this work differs. The wider setting is by now established: LLMs orchestrating chemistry tools plan and execute real syntheses (ChemCrow[7], Coscientist[8]), which is what makes the narrower question here worth asking — those systems *call* an elucidation routine, and this paper measures what such a routine can be expected to return. General-purpose LLMs have been applied off-the-shelf — Anthropic's forward/inverse white paper[2], SpectraLLM and SpecMol (multimodal LLMs over multi-spectral input)[9,10], and knowledge-enhanced tree-search reasoning[11] — and dedicated multimodal *benchmarks* now exist, most prominently **MolPuzzle**[12] (IR+MS+^{1}H+^{13}C elucidation puzzles with the molecular formula supplied). Closest to the present method, **IR-Agent**[13] introduces a multi-agent LLM framework that emulates expert IR interpretation and evaluates on experimental infrared spectra. These establish that the task is worth benchmarking, that LLMs can be scored on multi-spectral input, and that agentic decomposition of the interpretation helps; we claim priority on none of them. Our contribution is orthogonal to IR-Agent's: where it improves *how the model reads a spectrum*, we ask what limits the outcome once it has, and answer with a measured decomposition — generation recall versus verification precision — plus the blind benchmark and open dataset needed to measure either honestly. What is new here is three things that, to our knowledge, no prior benchmark combines. **(i) Real, literature-mined experimental spectra at scale:** prior multimodal benchmarks and trained baselines evaluate either on simulated or software-predicted spectra[1,6] or on *curated single-instrument* libraries — Alberts et al. use NIST gas-phase IR[14,15] — whereas IRSpectra-Bench is drawn from IRexp's experimental IR + ^{1}H + ^{13}C **as reported by the authors of the source papers**, across thousands of laboratories and instruments, for out-of-distribution compounds. The contrast we can draw is therefore between literature-reported heterogeneity and both simulation *and* curated uniformity, not between simulated and real alone. The closest trained precedent, Alberts et al.[14], learns an IR→structure transformer from ~635k *simulated* spectra with a 3,453-spectrum experimental fine-tune (top-1 44% on 6–13 heavy atoms), since improved by the same group to **63.8% top-1 / 84.0% top-10 on experimental NIST gas-phase spectra** given the formula, pretraining on 1,399,806 simulated spectra[15]. That line is the state of the art for *trained, formula-conditioned IR* elucidation. Its evaluation is restricted to **6–13 heavy atoms**, which is the size range where our own accuracy is highest — 60.5% top-1 for ≤15 heavy atoms (§4.1) — so on comparable molecules the training-free LLM and the purpose-trained transformer are closer than the headline numbers suggest; the gap our benchmark exposes opens up at the *larger* sizes their evaluation excludes (7.0% above 25 heavy atoms). IRexp instead releases experimental IR at scale as an open, redistributable resource and pairs it with a blind LLM benchmark. The two are complements rather than competitors — their models need exactly the kind of experimental data IRexp exists to supply, and neither their setting (single modality, gas-phase NIST, formula-constrained decoding) nor ours bounds the other. **(ii) Blind, fully specified, mechanical scoring:** reported accuracies on the same task vary enormously with inference method and scoring harness — the *same model* on the *same benchmark* spans a factor of forty. GPT-4o is reported at **1.4%** on MolPuzzle by the benchmark's own authors[12], at **27.8%** under a plain chain-of-thought harness by Zhuang et al., and at **57.8%** when the latter add knowledge-enhanced tree-search reasoning[11] — so numbers across papers are not directly comparable; we therefore fix and release a single, pre-registered, RDKit-InChIKey protocol with bootstrap CIs. **(iii) A recall/verification decomposition:** existing benchmarks report a single aggregate score, whereas we factor performance into generation recall and verification precision and show *where* the task is lost. That decomposition is the part we find stable under the perturbations we could run: four Claude models, a second chemical domain, and four different verifiers all stay recall-bound (§4.4, §4.5, §5.4). It is not tested outside the Claude family (§7).

Computational NMR for structure validation. Our forward-verification method is the LLM analog of a workflow chemists already trust: assign a structure by computing the spectrum each candidate *would* give and matching it to experiment. In solution, this is the DP4 / DP4+ probabilistic framework over GIAO-DFT shifts[16,17]; in the solid state it is **NMR crystallography**, where GIPAW-computed shifts adjudicate between candidate structures[18,19]. We replace the quantum-chemical predictor with a forward LLM, trading accuracy for zero setup cost, and inherit the same core principle — *verification by forward prediction is easier than inverse generation.*

The generate-and-forward-verify loop is not ours, and its best instantiation corroborates our diagnosis. NMR-Solver[3] builds the same loop without any LLM: it retrieves and fragment-recombines candidates, then ranks them by $^{1}H/^{13}C$ shifts forward-predicted with NMR-Net (^{13}C MAE 1.098 ppm) against the observed spectrum. On ~450 experimental literature spectra with the formula supplied it reports **52.89% top-1**, well above our 28.4%. We do not claim the loop as a contribution; §5 asks a different question — *how much of the remaining error is generation and how much is verification* — and answers it by decomposition (§5.2) rather than by building a better solver. The comparison is in fact the sharpest external evidence for our central mechanistic claim: NMR-Solver's predictor is roughly twice as sharp as the ~2 ppm LLM forward-predictor whose resolution §5.1 identifies as the binding constraint, and it converts that sharpness into roughly twice the top-1. That is what our §5.4 ablation predicts and what §5.1's separability measurement implies. **NMRAgent**[20] is the closest LLM-agent counterpart, coupling spectral tools to a knowledge graph and validating on newly isolated natural products; it is complementary to our aim, which is measurement of an off-the-shelf model rather than a best-effort system.

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2. The IRexp dataset

2.1 Motivation

What an IRexp record is. Each record is a **band list** — the peak positions in cm^{-1} that an author transcribed into the text of a paper (e.g. "IR (KBr): 3373, 3045, 2914, 1664 cm^{-1}") — together with the $^{1}H/^{13}C$ shift lists reported alongside it and, where resolvable, the compound's 2D structure. **IRexp contains no absorbance traces.** This is the form in which experimental IR is overwhelmingly *published*, and it is the form a language model consumes, but it is a different object from a digitised spectrum and the two should not be counted against one another.

Open experimental IR is scarce *in a redistributable, ML-ready form*, and the existing resources divide into two kinds that are not directly comparable.

Digitised spectra (absorbance traces). The largest freely downloadable collections are the NIST WebBook[21] (~16k IR spectra) and the Chemotion electronic-lab-notebook deposit[22] (~2k). The AIST SDBS[23] is larger (~54k FT-IR, all structure-linked) but is *view-only*: it caps downloads at 50 spectra/day, forbids commercial use, and ships no permissive licence or bulk export, so it cannot be used to train or redistribute open models at scale. Commercial libraries (e.g. Wiley KnowItAll, ~10^5 spectra) are larger still but closed. **On this axis IRexp is not the largest resource, and does not compete: SDBS alone holds more structure-linked spectra than IRexp holds structure-linked band lists.**

Band lists. Against text-derived peak-list resources, no large, permissively-licensed, structure-linked, bulk-reusable IR collection existed. IRexp fills that gap, and to our knowledge is the largest *openly redistributable* collection of experimental IR **band lists** by record count. The claim is scoped to that object type deliberately; the contribution is redistributable scale in the published-data modality, not a spectral library.

2.2 Construction

A per-compound IR band list, together with co-reported $^{1}H/^{13}C$ NMR, follows a remarkably stable textual convention in the experimental sections of organic chemistry papers. We exploit this with a browser-free harvesting agent:

- **Discovery.** Open-access primary literature is enumerated through the NCBI E-utilities and harvested in bulk from the PMC Open-Access Subset[24] on AWS S3 (plain HTTPS, no anti-bot, fully redistributable CC-BY content), supplemented by the Chemotion FT-IR deposit (RADAR4Chem, CC-BY-SA-4.0).
- **Extraction.** A deterministic parser segments experimental text into per-compound records and extracts IR wavenumbers and $^{1}H/^{13}C$ shift lists, with quality gates that reject instrument scan-range artefacts and prose false-positives (band-list density, $\geq$4 bands, plausible 400–4000 cm^{-1} window).
- **Structure resolution.** In-text IUPAC names are converted to SMILES with OPSIN[25], canonicalised with RDKit[26] (InChIKey, SELFIES[27]), with a PubChem[28] fallback for trivial/natural-product names. A key engineering finding was that the dominant open-access main-text convention labels compounds with *letter-prefixed* series labels (e.g. "…carbothioamide **(B1)**:") rather than the digit-first "(3a)" of supporting-information sections; capturing these and cleaning narrative/PDF artefacts before resolution raised structure coverage from 24% to **35%**.

The pipeline is browser-free by design. This is not an anti-bot bypass — the bulk corpus is fully open — but a throughput choice: plain HTTP from a bulk corpus parses at hundreds of papers per second, which is what makes a 10^5-scale harvest feasible where a per-page browser driver (as used to build prior 10^3-scale sets) is not.

2.3 Contents and licensing

Table 1. IRexp dataset contents and provenance.

field	value
experimental IR band-list records	**121,233**
…co-reporting ^{1}H and/or ^{13}C NMR	87,075 (72%)
…with a resolved 2D structure	**43,060 (35.5%)**
…of these, with ^{1}H and/or ^{13}C NMR	**40,702**
…of these, with both ^{1}H and ^{13}C (full quadruples)	**33,201**
by provenance: author-transcribed from PMC-OA text (CC-BY)	119,345
by provenance: peak-picked from Chemotion ELN deposits (CC-BY-SA)	1,888

Both pools are stored in the same band-list form, but they are not the same object and we keep them separable. The PMC pool is transcribed from the experimental section of a paper and carries a median of 9 bands per record; the Chemotion pool derives from deposited spectra and, being peak-picked rather than author-selected, carries a median of 39. Users training on band density should treat the pools separately. The discriminator is each record's `source_doi`: Chemotion records carry the RADAR4Chem prefix `10.22000`, PMC records a `PMC:` accession. The released file has no separate `license` column, so `scripts/split_license_pools.py` performs the split and stamps each record with its licence.

Extraction fidelity is measured, not assumed. Because every record cites its source accession and PMC is open, transcription can be checked directly. On a seed-fixed random sample of 60 PMC-sourced records (`scripts/audit_extraction.py --n 60 --seed 0`), we re-fetched each source article and asked whether every recorded wavenumber appears in that article's text: **560/560 bands and 60/60 records were confirmed** (Wilson 95% CI 99.3–100% and 94–100% respectively). This bounds *transcription* error — hallucinated, mis-parsed or unit-mangled values — at under 1% of bands. It does **not** measure whether the parser found every IR string in every paper, which is a recall question that requires human reading; that manual audit is prepared but not yet run (§7). The audit record is released at `data/audit/extraction_audit.json`.

A structure-complete split, **`irexp_resolved`** (43,060 records, 100% structure-linked), is the training-/benchmark-ready subset and is ~6× the 6,833-molecule set used to train Spectro[1] (Fig. S2). Provenance is 119,345 PMC-OA records (CC-BY) plus 1,888 Chemotion records (CC-BY-SA); the two licences are kept as separable pools. Each record is DOI-/accession-traceable. Re-resolution is additive and content-keyed, so the dataset can be re-enriched without re-mining.

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3. Benchmark design (IRSpectra-Bench)

From `irexp_resolved` we draw **IRSpectra-Bench**, 194 blind elucidation problems. Each problem presents the **molecular formula** (as from HRMS), the **IR band list**, and the **^{1}H and ^{13}C shift lists** (with multiplicities and J-couplings where reported), and asks for the structure. No name, SMILES, or hint is given. Every main-round ground-truth structure is **spectrally validated** by an automated RDKit consistency check (^{13}C peak count vs symmetry-unique carbons, molecular-formula match, SELFIES round-trip), excluding records with merged or incomplete spectra (6/140 in the main round, leaving 134). The exclusions are itemised rather than asserted: five report more ^{13}C peaks than the structure has carbons (R03 28>24, R14 11>8, R65 26>18, R67 23>16, R131 34>17 — merged or contaminated spectra) and one is too sparse to constrain (R82, 5 peaks for 22 symmetry-unique carbons). The filter is deterministic and runs over all three rounds in `scripts/validate_benchmark.py`, which regenerates every `clean_qids.json` from the released questions and answers, so the cohort behind the headline number is reproducible rather than a shipped artifact.

The filter tests ^{13}C against the carbon count but does not gate on ^{1}H, and we report what that leaves. In **13 of the 194** retained records the total reported ^{1}H integral exceeds the hydrogen count of the reference structure — extra signal from residual solvent, water, exchangeable protons or a reported rotamer mixture. The audit prints these as a diagnostic rather than excluding them, because the cohort was fixed in advance and re-filtering it post hoc on a criterion chosen after seeing the results is precisely the degree of freedom a benchmark should not take. Instead we report the sensitivity: dropping all 13 moves the headline from **28.4% to 29.3%** (53/181), a shift of +0.9 points that leaves every conclusion in this paper unchanged. Readers who prefer the stricter cohort can regenerate it from the diagnostic. The 60 compounds of the controlled rounds are retained as fixed, pre-registered sets for the difficulty and within-compound controls, with the same self-consistency audit reported separately rather than used to exclude (57/60 pass; §7). Problems are stratified by RDKit ring analysis: a compound is **simple** iff it has at most two rings (single ring or two separate ring fragments), no fused/spiro/bridgehead system, and $\leq$22 heavy atoms; every other compound — any fused/spiro/bridged or $\geq$3-ring system, **or** more than 22 heavy atoms — is **complex** (98 simple / 96 complex). This binary difficulty axis is distinct from the continuous size gradient in §4.1, which bins by heavy-atom count ($\leq$15 / 16–25 / >25). The criterion is therefore exhaustive (the 23–24-atom band, 13 compounds, is classed complex by size). The 22-atom threshold is not load-bearing: sweeping it from 18 to 26 moves the simple-minus-complex top-1 gap only between 36 and 40 points (39.6 at the released value), so the separation is a property of the compounds rather than of where the line was drawn (`scripts/difficulty_sensitivity.py`). InChIKey de-duplication is applied across all rounds to prevent leakage.

Scoring is mechanical. A prediction is *correct* if its RDKit InChIKey connectivity layer (first 14 characters) matches the reference — i.e. we score *constitution* (atom-and-bond connectivity); stereochemistry is reported separately. This is a deliberate floor: a candidate with the correct constitution but wrong stereochemistry counts as correct. We report the strict alternative rather than assert it is immaterial. Scoring the *full* InChIKey (stereochemistry-sensitive) gives **21.1% top-1 and 25.8% recovered** (41/194 and 50/194), against 28.4% / 33.5% at the connectivity layer — 7.3 points lower, so the choice of layer is not immaterial and the constitution figure should be read as an upper bound on full-stereochemistry accuracy. We nevertheless take constitution as the headline metric because 1D ^{1}H/^{13}C/IR rarely fixes absolute configuration, so a stereochemistry-aware score is not well-posed from the given data: only 10.3% (20/194) of benchmark answers carry a *defined* (assigned R/S) stereocentre, and a model is penalised there for information the prompt never contained. Both numbers are produced by `scripts/score_main.py --stereo`.

Metrics. We report five quantities. **Top-1 (exact constitution)** is the fraction of compounds whose single best-ranked candidate matches the reference at the InChIKey connectivity layer. **Recovered (top-3)** is the fraction for which the reference appears among the up-to-three ranked candidates returned (matching the lenient "recovery" protocol of ref.[2]). Where this paper writes *recovery* it is always reporting another study's metric under that study's own name; our own quantities are always named **recovered (top-3)**, **generation recall** and **conditional-on-recall precision**. **Generation recall** is the fraction for which the reference is present in the candidate pool *before* re-ranking — the ceiling any verifier can reach. Recovered (top-3) and generation recall share the same denominator (all compounds) and differ only in which candidate set is searched: the up-to-three returned, versus the full pool. They therefore coincide wherever the pool is the returned three, which is everywhere except the generate-wide arm of §5.3, where the pool is larger. The denominator changes only for conditional-on-recall precision, which is taken over recall-positive compounds alone. **Conditional-on-recall precision** is the verifier's hit rate over recall-positive compounds only, isolating verification quality from generation. The forward verifier ranks candidates by a symmetric **chamfer distance** between predicted and observed ^{13}C peak sets (for each predicted peak, the distance to its nearest observed peak, summed, and symmetrically for each observed peak; lower is better, with no equal-count requirement). We also report Morgan(2, 2048)[29] Tanimoto as a graded "right scaffold/family" signal. CIs are bootstrap 95% over compounds; model-vs-model differences use McNemar's exact test with Holm correction.

Solvers are LLM agents run under a consumer subscription. A frontier LLM (Claude Opus) is invoked as an independent sub-agent per batch of problems; agents are closed-book — configured with no web/tool access beyond an RDKit formula check and no access to ground truth, verified by grep-auditing their task transcripts at run time (audit logs available on request; the committed artefacts are the parsed per-compound predictions). Adherence to the supplied formula is high but imperfect: 91.3% (115/126) of forward-verification candidates match the given molecular formula exactly (100% of true structures, 89.6% of decoys), and 76.6% across the full top-3 pool — a residual generator error orthogonal to regiochemistry.

That adherence is **not uniform across rounds**, and the two figures above come from the better-behaved ones, so we give the breakdown rather than let a reader carry either number across. On top-1 answers it is 95.0% (38/40) in the v3 round and 90.0% (18/20) in the v2-control, but **77.6% (104/134) in the headline main round** — which is to say the solver returns a structure of the wrong composition, in violation of a constraint it was explicitly handed, for about one main-round compound in five. Those answers are scored as misses like any other, so this inflates nothing; it does mean the formula check described above was not equally effective in every round, and the main-round arm should be read as the weakest-constrained one. `scripts/analyze_misses.py` regenerates the breakdown. This makes the benchmark free to run and reproducible without API credits.

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4. How well do LLMs elucidate real structures?

4.1 Headline performance

Solver agents work blind from formula + IR + ^{1}H + ^{13}C and return up to three ranked candidates. Each agent handles a small batch of problems in a bounded context that is reset between batches (2–12 compounds per context in the released run; see `data/benchmark_main/raw/`), rather than one long context over the whole benchmark — the arm §4.3 shows matters. Over the full **194-compound benchmark** (134 spectrally-validated compounds + the 60 from the controlled rounds), with bootstrap 95% confidence intervals:

Table 2. Headline elucidation performance on IRSpectra-Bench (n=194), overall and by difficulty stratum, with bootstrap 95% CIs.

metric	overall (n=194)	simple (n=98)	complex (n=96)
top-1 exact constitution	**28.4%** [22–35]	48.0% [39–57]	8.3% [3–15]
recovered (within top-3)	33.5% [27–40]	54.1% [44–63]	12.5% [6–20]
scaffold-level (best Tanimoto $\geq$ 0.45)	56%	73%	39%
mean best Tanimoto	0.59	0.73	0.45

As a cohort-robustness check, restricting the controlled rounds to their spectrally-clean subsets as well (134 main-clean + 57 controlled-clean = 191 strictly-validated compounds) leaves the headline unchanged: top-1 **28.8%** (95% CI 23–36), recall 34.0%, with simple 49.0% / complex 8.4% — within ~1 point of the n=194 figures on every metric shown, with fully overlapping intervals, so the asymmetric inclusion of the pre-registered controlled sets does not drive the result.

The gradient is sharp and the intervals are tight. The 48%→8% simple→complex separation (Fig. 2) confirms the benchmark is discriminating across a realistic difficulty range, and accuracy falls monotonically with molecular size in step with it — top-1 **60.5%** for ≤15 heavy atoms, 28.3%

for 16–25, and **7.0%** above 25 (Fig. S3). The model recovers molecular formula and functional groups reliably and the scaffold often (best Tanimoto ≥ 0.45 for 56% of compounds), but the exact constitution far less often.

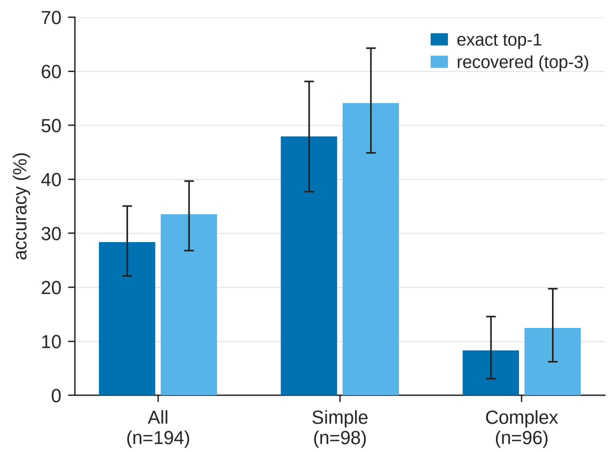

Fig. 2: Top-1 and recovered accuracy on IRSpectra-Bench by difficulty (all / simple / complex, n=194), with bootstrap 95% confidence intervals. The benchmark separates a realistic difficulty range: simple targets are solved four to six times as often as complex ones on both metrics.

What a failure actually looks like is measurable rather than a matter of impression, so we measured it over all 139 top-1 misses (`scripts/analyze_misses.py`). **76.6% are constitutional isomers of the true structure** — exactly the right atoms, assembled wrongly — against 23.4% with the wrong molecular formula outright. The composition is usually right; the connectivity is not.

The narrower reading often given to this result is not supported. Only **22.6%** of misses share the true Murcko scaffold, i.e. are genuine positional errors of the kind "*which* ring position the substituent occupies", and just 2.9% reach Tanimoto ≥ 0.85; the median Tanimoto between an isomeric miss and the truth is 0.39. So regiochemistry in the strict sense accounts for roughly a fifth of failures, not the bulk of them — most isomeric misses are more substantial rearrangements of the same atoms. Both facts point the same way about 1D data (near-degenerate $^{1}H/^{13}C$ shifts under-determine connectivity, which is why 2D experiments such as HMBC and NOESY exist), but the honest statement is *wrong connectivity at the right composition*, with strict regiochemistry a well-defined minority of it.

4.2 Reconciling with prior reports

Our 28% top-1 sits far below the ~100% on "simple" molecules reported, in a non-peer-reviewed company white paper, for the same model class.[2] Because that figure has not been independently scored, we reconcile it against the *peer-reviewed* record (the MolPuzzle benchmark[12] and its re-scorings[11], and the trained baselines[1,6]) rather than treating it as settled. Three methodology and scoring choices differ between that report and ours, each in the direction that would raise a reported number; we can name them but not apportion the gap between them, so we do not claim a decomposition:

- **Difficulty.** "Single ring" by ring-count is not "easy": our simple stratum includes, e.g., a hexasubstituted benzene whose regiochemistry has many realisations. A low ring count is therefore no guarantee of an easy problem — though a high one is a strong difficulty signal, since ≥4 rings appear in 38% of recall-negative compounds against 3% of recall-positive (§5.3). Ring count bounds difficulty from below, it does not proxy for it.

- **Scoring.** Prior work counts a recovery if the reference appears among three ranked candidates over three independent runs; we report single-run top-1 and top-3.
- **Hints.** Prior hard targets received the starting-material SMILES, which fixes most of the scaffold; we give no starting material. We do supply the molecular formula, which is itself a real constraint — §4.6 measures what it is worth alone (5% top-1 with the spectra masked) — so our setting is *less* hinted than that comparison but more hinted than formula-free generative baselines (§4.2 above).
- **Curation.** Prior compounds were hand-selected; ours are scraped and unfiltered for solvability.

On a like-for-like easy/hinted/lenient setup the numbers rise; on the realistic, hint-free, scraped regime, ~28% is the honest figure.

Versus trained models — a bound, not a leaderboard. A like-for-like comparison against specialised trained models is not available: no system has been scored on an identical test set, and published numbers differ in the three respects that most move the score — spectrum realism (simulated/curated vs. real), hints, and how "exact match" is defined. Most trained baselines report their accuracy *in-distribution on simulated spectra*, though not all: Alberts et al. reach 63.8% top-1 on **experimental** NIST gas-phase IR with the formula supplied[15], a genuinely real-spectrum result we do not discount — its test set is a curated single-instrument gas-phase library of **6–13-heavy-atom** molecules rather than heterogeneous literature-reported band lists, and its input is IR alone, but it is experimental data and the number stands. The two evaluations barely overlap: our compounds span 8–60 heavy atoms (median 20) and only **15 of 194 (8%)** fall inside their 6–13 window. Against our own ≤15-heavy-atom stratum (60.5%) their 63.8% is a near-tie; the divergence is a property of molecular size, which is exactly what §4.1 measures and what an evaluation capped at 13 heavy atoms cannot see. Of the multimodal systems: Spectro (^{1}H/^{13}C/IR→SELFIES, 6,833 molecules) reports **93%** overall test accuracy trained jointly with its IR vision model and **82%** with fixed embeddings — but on a 1,366-molecule held-out split whose IR is plotted from reference data and whose NMR is software-*predicted*, not experimental[1]; and NMIRacle, which conditions jointly on IR+^{1}H+^{13}C, reports 48% top-1 / 66% top-15 exact-SMILES recovery — again on held-out molecules from a *simulated* corpus drawn from the training distribution (an 8:1:1 split of the ~790k-molecule simulated set of Alberts et al.), a limitation its authors state themselves[6]. We measure 28.4% top-1 (33.5% top-3) on the full n=194 benchmark — rising to 29.9% with forward verification over that same full benchmark (§5) — on **blind, real, literature-mined experimental** spectra of out-of-distribution compounds.

One asymmetry runs the other way and must be stated, because it cuts against us. NMIRacle takes **no molecular formula**; its authors explicitly count "assumptions of strong prior information, such as chemical formula or molecular scaffold" among the limitations of prior work. We *do* supply the formula (§3), which is a substantial constraint — it fixes composition and, as §4.6 shows, alone accounts for 5% top-1. So the two settings differ on two axes at once and in opposite directions: our spectra are real and out-of-distribution where theirs are simulated and in-distribution, but their input is strictly harder than ours. Neither number bounds the other. These are not comparable as a leaderboard; read only as a *bound on the simulated-to-real gap*, the contrast suggests that high in-distribution accuracies substantially overstate real-world performance — the same gap we document for the LLM above. The instability of the metric itself reinforces the caution: the same

~40× MolPuzzle swing documented in §1.1 (GPT-4o: 1.4%[12] to 27.8% to 57.8%[11], method- and harness-dependent) bounds how much weight any single unaudited near-100% claim[2] can bear.

4.3 Methodology dominates: a within-compound control

The same 20 molecules were solved two ways: (a) by a single LLM context handling all of them sequentially with no tools, and (b) by four independent agents of five compounds each, with RDKit formula-checking — i.e. bounded, reset contexts. On the *identical* compounds, **recovered (top-3)** rose from **5% to 15%** (1/20 → 3/20) and top-1 from 0% to 15% (0/20 → 3/20), with zero sample confound. Because §3 pins McNemar's exact test for paired comparisons we apply it here too, and it does not reach significance: the top-1 arm is necessarily nested (arm (a) solved none) at b=0, c=3, giving **p=0.25**, and the recovered arm reaches at best **p=0.5**. The 15% point estimate carries a Wilson 95% CI of roughly 5–36%. This is therefore a directional within-compound demonstration, not an established multiplier, and we do not claim the 3× as a measured effect size. Small rounds also swing widely (15–40% across n=20–40 draws), which is why the headline is the full **194-compound** figure (28.4% top-1, 95% CI 22–35) rather than any single round. The practical lesson has to be stated at the strength the evidence supports, which is weaker than the point estimate invites: bounded, frequently-reset contexts with tool access appear to raise measured performance, in a direction consistent across both arms but **not established in size** at n=20 (p=0.25). Read that way it still plausibly explains *part* of the gap to optimistic prior reports, whose per-problem API calls implicitly used method (b) — but it is a hypothesis about that gap, not a quantified contribution to it.

4.4 Model comparison: the benchmark ranks capability (and separates the extremes)

A benchmark is only useful if it separates models. On a fixed 24-compound subset solved blind by four Claude models — spanning a wide capability range, including the newest (Fable 5) — under the identical protocol (Fig. 3):

One asymmetry must be disclosed, because §4.3 shows the variable it concerns has a large effect. The three comparison models each solved the 24 compounds as four six-compound contexts, whereas the Opus column reuses the headline run, in which those same 24 items were packed as one six-compound and two twelve-compound contexts. Context packing is exactly the factor §4.3 measures at 5%→15%, so the Opus point estimate is not strictly protocol-matched to the other three and, if anything, is the arm handicapped by longer contexts. This does not affect the nesting or the Fable-vs-Haiku contrast, but a clean re-run of the Opus arm under four six-compound contexts is the right fix and is listed as outstanding in `docs/MODELS.md`.

Table 3. Four-model comparison on a fixed 24-compound subset under the identical blind protocol.

model	top-1	recovered	top-1 95% CI
Claude Fable 5	**45%**	54%	[25, 67]
Claude Opus	25%	29%	[8, 42]
Claude Sonnet	20%	25%	[8, 38]
Claude Haiku	**0%**	4%	[0, 14]

CIs are bootstrap except Haiku's, which is the Clopper–Pearson exact interval for 0/24 (the percentile bootstrap is degenerate at a boundary count of zero).

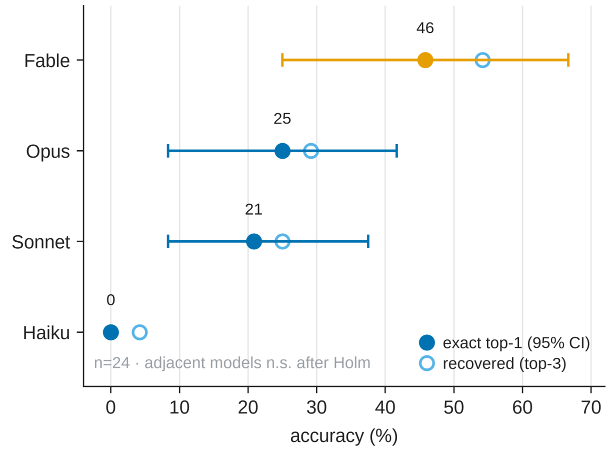

Fig. 3: Four-model comparison on a fixed 24-compound subset, solved blind under one protocol: Fable 5 45% > Opus 25% > Sonnet 20% > Haiku 0% top-1. The outcomes are strictly nested — each stronger model solves a superset of the weaker one's compounds — so the benchmark is capability-sensitive, but at n=24 it is underpowered to separate adjacent models (§4.4).

Three signals matter here. First, the four models rank in **monotonic capability order**, and the outcomes are **strictly nested** (Haiku $\subset$ Sonnet $\subset$ Opus $\subset$ Fable — each stronger model solves a superset of the compounds the weaker one does), with the smallest **floored at 0% exact** (4% recovered) on the same problems — so the benchmark is **capability-sensitive** and not a coin-flip. The nesting makes the *ranking* robust, but at n=24 the subset is **underpowered to separate adjacent models**: by McNemar's exact test only the Fable-vs-Haiku gap survives multiple-comparison correction (Holm-adjusted p=0.006); the Fable-vs-Opus gap is not significant (uncorrected p=0.063, Holm-adjusted p=0.19), and Opus vs Sonnet are statistically indistinguishable — the bootstrap CIs above overlap accordingly. Second, two mid-tier frontier models agree closely (Opus 25%, Sonnet 20%), so the recall-bound regime of §4.1 (top-1 in the high-20s%) is **not an artefact of a single model**. Third, the newest model nearly **doubles** the next-best top-1 (45% vs 25% on identical compounds; a large but, at this n, not-yet-significant gap) yet still misses the majority — IRSpectra-Bench is **hard and far from saturated even for the strongest model available**, with clear headroom. We ran four Claude-family models because they are callable for free under one subscription; a true cross-vendor sweep (GPT-, Gemini-class, open models) needs API access we deliberately did without, but the monotonic, large capability spread makes it unlikely the recall-bound pattern is specific to one model lineage. This is the behaviour a benchmark needs to remain informative as models improve.

4.5 Domain case study: battery-electrolyte chemistry

Structure elucidation from spectra is the daily inverse problem of the electrolyte-and-interphase community: the soluble decomposition products of carbonate and sulfone electrolytes, the additives that build the solid-electrolyte interphase, and the fluorophosphate/fluorosulfonyl species that NMR is routinely used to assign are exactly the molecules whose constitution must be read back out of IR and multinuclear NMR. To test whether the recall-bound regime above holds for *this* chemistry — rather than for organic molecules at large — we curated **IRSpectra-Bench-Electrolyte**, a 48-compound subset (eight per class as curated; two later yielded no parseable candidate, leaving **46**

scored) of structure-resolved IRexp records selected by substructure for the six functional families that dominate lithium- and sodium-battery electrolytes and their breakdown products: **carbonate** (linear/cyclic, the EC/DMC backbone), **sulfonyl/sulfonate** (sulfones, sulfonamides, the $-SO_2CF_3$ motif of imide salts), **nitrile** (acetonitrile- and adiponitrile-type high-voltage additives), **sp^3-C–F** (fluorinated solvents and additives), **phosphoryl** (phosphates/phosphonates, the $LiPF_6$-derived OPF chemistry), and **glyme/oligoether** (the ether solvents and PEG linkers). These are literature compounds bearing the *functional chemistry* of electrolytes, drawn from the open corpus; they are **not** operando or in-cell degradation spectra, and we make no claim that they are. They are excluded from every other split in this paper. Each compound was solved blind under the identical decoupled-agent protocol (Opus, closed-book, up to three ranked candidates, RDKit only for formula/parse checks).

Performance lands in the **same broad regime as the headline benchmark** — overall **top-1 26%, recovered 28%** (12/46 and 13/46; two compounds received no parseable candidate; at n=46 the 95% CI is wide and overlaps the headline) — consistent with the bottleneck being a property of the elucidation task rather than of any one chemical neighbourhood. The recall-bound *signature* is present in the same form: the true structure enters the candidate set for only 13 of 46 compounds, and of those 13 the solver already ranks it first in 12, so ranking is not what limits this subset either. We did **not** run forward-verification here, so this reproduces the recall-bound pattern under the solver's own ranking; it does not independently reproduce the §5 verification-precision measurement. The per-class breakdown (Fig. S4) is itself informative:

Table 4. Per-class performance on IRSpectra-Bench-Electrolyte (n=46).

electrolyte class	n	top-1	95% CI	recovered (top-3)	95% CI
sp^3-C–F	8	**50%**	[22, 78]	50%	[22, 78]
carbonate	7	29%	[8, 64]	43%	[16, 75]
phosphoryl	8	25%	[7, 59]	25%	[7, 59]
glyme / oligoether	7	29%	[8, 64]	29%	[8, 64]
sulfonyl / sulfonate	8	**12%**	[2, 47]	12%	[2, 47]
nitrile	8	**12%**	[2, 47]	12%	[2, 47]

Intervals are Wilson score intervals for a binomial proportion, appropriate at these counts where the percentile bootstrap is degenerate. They overlap almost completely.

At n=7–8 per class these per-class differences are within sampling noise (six-class χ^2 p≈0.56; best-vs-worst Fisher exact p≈0.28), so the ordering is a hypothesis-generating pattern rather than an established ranking. Read that way, it is chemically legible. **sp^3-C–F** centres are the easiest: a C–F coupling fingerprint (the large $^1J/^2J$(C,F) splittings) localises the fluorine and pins regiochemistry, removing exactly the degeneracy that defeats elucidation elsewhere. **Sulfonyl** and **nitrile** are the hardest, for two distinct reasons that recur in real electrolyte analysis: sulfur/phosphorus **oxidation-state ambiguity** — sulfide vs. sulfoxide vs. sulfone, $-SCF_3$ vs. $-SO_2CF_3$, the very distinctions an interphase study must make — is poorly constrained by $^1H/^{13}C$ shifts alone (it lives in the IR and in heteronuclei the benchmark does not score); and nitrile-bearing targets in the corpus sit on heavily substituted heteroaromatic cores whose regiochemistry the 1D spectra underdetermine. This is the recall-bound failure of §4.1 reappearing in a domain-specific guise, and it is precisely where the **forward-verification** recipe of §5 — compute the spectrum each candidate *would* give and match it to the one observed — plays a role *analogous* to (not a replacement for) computational-NMR / NMR-crystallography structure validation, sharing its forward-prediction logic while trading DFT's calibrated accuracy for zero setup cost. The subset, its per-class answers, and the scorer are released with the benchmark.

4.6 Is the model reading the spectra? A formula-only control

Because every benchmark compound is mined from open-access literature, a frontier model may have encountered it during pretraining. Benchmark contamination of this kind is a general and well-documented hazard for LLM evaluation,[30] and it is the objection any literature-mined benchmark must answer first. The cheapest decisive test of whether it explains the headline number is to take the spectra away. We ran the identical blind protocol with every spectral channel masked, so the solver receives the molecular formula and nothing else (`scripts/modality_ablation.py`, condition `formulaonly`; solvers were barred from reading any repository file or searching the web, and RDKit was permitted only to check that a proposed SMILES parses and matches the formula). A molecular formula does not determine constitution, so accuracy materially above the floor would indicate recall rather than reasoning.

One asymmetry must be disclosed, because a companion experiment shows this exact structure can manufacture a large spurious effect. Only the **formula-only** arm was freshly generated (2026-07-28); the **full-modality** comparison arm re-uses the archived June predictions — all 60 of its top-1 answers are byte-identical to that run. An earlier leave-one-out attempt built the same way (fresh ablated arm against archived control) produced −IR *beating* full modality by 40 points, which is impossible and was discarded (`docs/MODALITY_ABLATION.md`). The reason that failure does not impugn this control is the **direction**: fresh agents reasoned harder per compound than the archived batched run, so the bias runs *toward* the formula-only arm, and it still collapsed to 5%. A confound that works against the finding cannot manufacture it. We nonetheless flag it, and the rule the discarded arm yielded — every condition must be generated in one campaign, control included — is why no leave-one-out modality result appears in this paper.

Table 5. Formula-only control, paired on the same 60 compounds as §5.

condition	top-1	recovered (top-3)
formula only	**3/60 (5%)**	3/60 (5%)
formula + IR + ^{1}H + ^{13}C	**14/60 (23%)**	19/60 (32%)

The outcomes are perfectly nested: **eleven** compounds are solved with the spectra and not without, and **none** the other way round (McNemar exact p=0.001). Removing the spectra removes the result.

A second, independent control: publication recency. The formula-only arm shows the spectra carry the signal, but not whether memorisation contributes to the part the spectra explain. A model can only have memorised a compound whose source paper was in its training corpus, and older papers have had longer in more corpora — so if recall drove the headline number, accuracy should fall with publication recency. We resolved the publication year of the source paper for **all 194** benchmark compounds from their accessions (`scripts/contamination_recency.py`); they span 2008–2026. Accuracy is flat: **28.6%** for the older half (≤2020, n=112) against **28.0%** for the newer (n=82), a point-biserial correlation between publication year and correctness of $\mathbf{r = -0.007}$, and no monotone trend across year buckets (Fig. 4b). The most recent bucket (≥2024, n=25) is in fact the highest at 40% [23, 59].

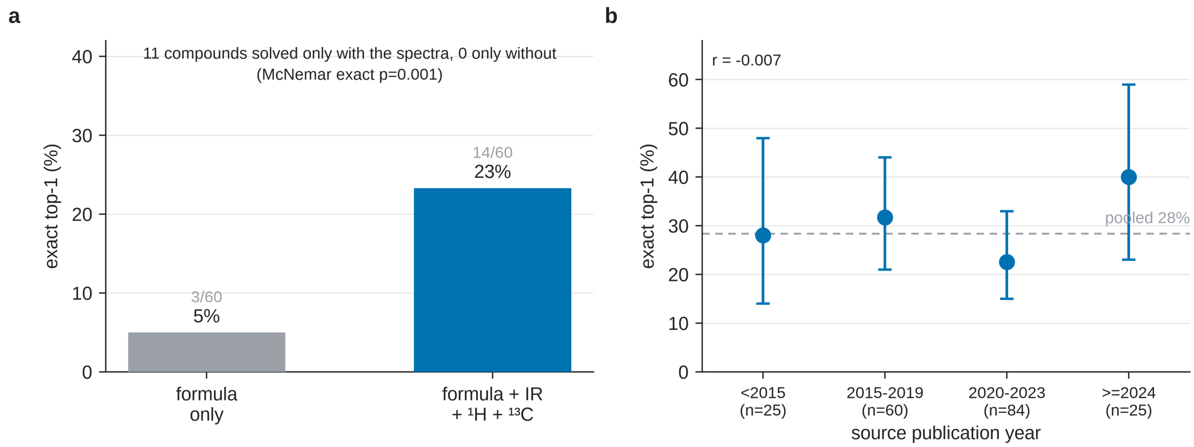

Fig. 4: Two contamination controls. (**a**) Removing the spectra: with only the molecular formula the solver reaches 3/60, against 14/60 with IR + ^{1}H + ^{13}C on the same compounds. The outcomes are nested — 11 compounds are solved only with the spectra and none only without (McNemar exact p=0.001). (**b**) Accuracy against the publication year of the source paper (n=194, Wilson 95% CIs): flat, point-biserial r = −0.007. Recall out of pretraining would predict a decline with recency; none is observed.

The raw split is if anything biased *against* the newer half, because newer papers skew to larger molecules (median 22 heavy atoms against 20) and size is the dominant driver of lower accuracy (§4.1). Stratifying by heavy-atom band removes that confound. Newer compounds lead in the two bands that carry most of the accuracy (≤15: 64% vs 58%; 16–25: 34% vs 25%) and trail slightly in the largest band, where both are near the floor (>25: 6% vs 8%). The size-adjusted older-minus-newer difference is **−5.1 points, 95% CI [−17.2, +7.0]** (Cochran–Mantel–Haenszel χ^2=0.42, p=0.51, continuity-corrected). Its point estimate has the opposite sign from what recall out of pretraining would produce, but the interval comfortably includes zero — so this **bounds** any recency effect rather than demonstrating a reversed one, and by the same standard §5.3 applies to its own adjacent conditions we do not read it as directional.

The 5% is not zero, and the three compounds are worth naming rather than rounding away — individually, they do not all support the same reading:

- $\mathbf{C_{15}H_{15}ClF_2O_3SSi}$, a 2-(trimethylsilyl)aryl sulfonate. The Si/S/Cl/F_2 composition is a near-unique benzyne-precursor signature and the compound is **absent from PubChem**, so recognition is implausible: here the formula really is close to determining.
- $\mathbf{C_{13}H_{19}NO_4S}$, *N*-tosyl-leucine (CAS 67368-40-5). The formula strongly suggests tosyl + leucine, but the compound is also catalogued and common; inference and recall are both available.
- $\mathbf{C_{17}H_{26}O_3}$, **[6]-paradol** (CAS 27113-22-0), a ginger natural product. This formula constrains very little by itself — a named natural product recovered from composition alone is more plausibly recognised than deduced.

So one of the three is clean chemical inference and one or two are consistent with memorisation. We say so because the arm exists to make such cases visible, not to argue them away, and because the bound is unaffected either way: formula-level recall accounts for at most about a fifth of the headline accuracy on this set. Together with the recency result, the two controls are independent and agree. Neither is a randomised experiment — publication year is observational and the formula-only arm cannot distinguish memorisation from inference on a near-determining formula — so we claim a strong bound rather than exclusion (§7). The control record is released at `data/modality/formulaonly_control.json`.

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5. Forward-verification elucidation

5.1 Method

The inverse direction is the model's hard, isomer-blind direction; the **forward** direction (structure→spectrum) is its easy, accurate one.[2] Regioisomers, crucially, have *different* forward-predicted ^{13}C shifts — though, as we quantify at the end of this subsection, by a margin far thinner than that framing suggests. We therefore close a generator–verifier loop:

> **generate** candidate structures (inverse) → **forward-predict** each candidate's ^{13}C spectrum, blind to the observed spectrum → **re-rank** candidates by the distance between predicted and observed ^{13}C → return the best match.

This mirrors how a chemist confirms a structure ("if it were that isomer, C-3 would be at ~120 ppm; we observe 135, so it is the other"), exploits the standard principle that verification is easier than generation, and requires no training. **Fig. 5** shows the mechanism on a real benchmark pair — the picolinamide / nicotinamide regioisomers, which the inverse task cannot separate but whose forward-predicted ^{13}C spectra match the observed one at 0.42 vs 1.30 ppm.

We implemented the verifier as independent LLM agents that predict ^{13}C shift lists from SMILES alone, using pure reasoning with no tools. All candidates from all compounds were pooled, canonicalised, shuffled and anonymised, then split into fixed-size batches, so a predictor sees neither the observed spectrum, nor the target's identity, nor which candidates belong to the same target. Batching does not separate a target's own candidates — in the §5.2 arm, 7 of 8 batches contained two candidates for some one compound — but because the predictor never sees an observed spectrum, co-occurrence carries no information about which candidate is correct; it can at most let the predictor notice that two structures are isomers, which is already evident from either alone. Predicted and observed ^{13}C peak sets were then compared with a symmetric chamfer distance.

For balance — and because conditional precision is 72–89%, not 100% — the verifier also produces clear **false positives** on near-degenerate pairs. On the 2-(nitrophenyl)-2,3-dihydroquinazolin-4(1*H*)-one targets ($C_{14}H_{11}N_3O_3$), the forward predictor cannot separate the *ortho*- and *meta*-nitrophenyl regioisomers: their predicted ^{13}C lie within its ~2 ppm error, and it ranks the wrong (2-nitrophenyl) isomer first by a chamfer of 1.35 ppm versus 1.36 ppm for the true (3-nitrophenyl) one — a 0.01 ppm margin, i.e. effectively a coin flip. This is the precision-loss mechanism of §5.3 made concrete: when candidates are near-degenerate the cheap predictor cannot resolve them, and it is exactly these cases — not generation failures — that a sharper (DFT-level) or 2D-NMR-grounded verifier would need to fix.

How thin is the margin the method works on? The premise above deserves a distribution rather than a worked example, so we measured the chamfer between the *predicted* spectra of every pair of candidates proposed for the same target (`scripts/isomer_separability.py`). For isomeric pairs — the regioisomer-like ones the verifier exists to separate — the median separation is **1.21**

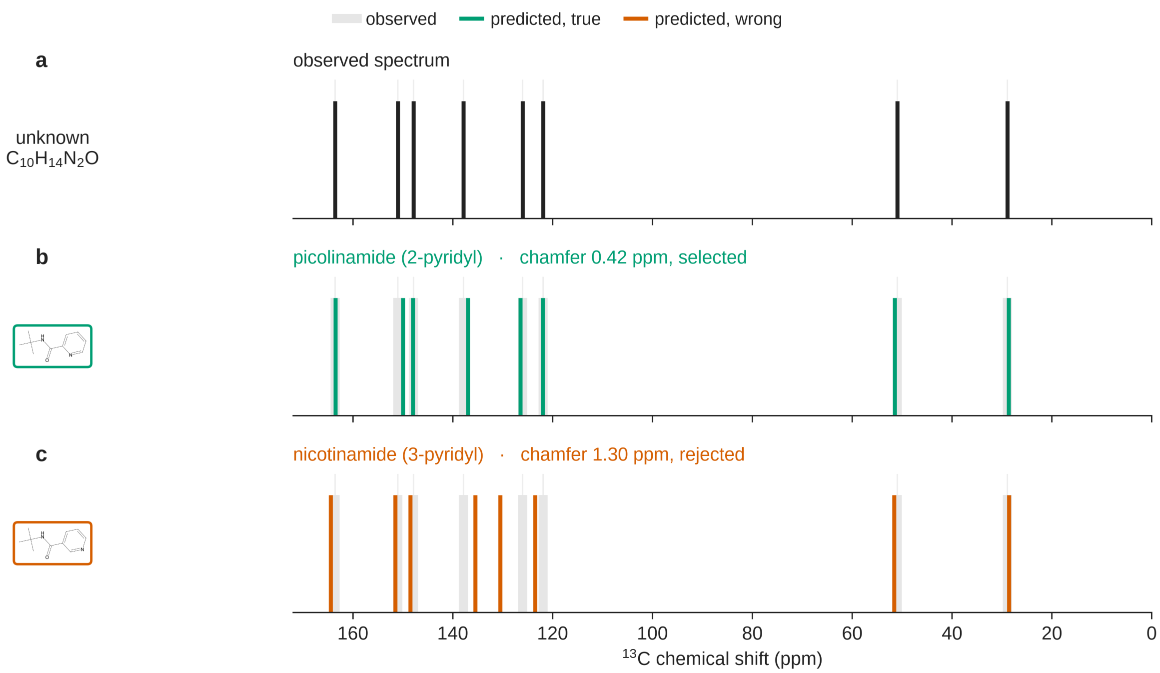

Fig. 5: Forward-verification on a real benchmark regioisomer pair. Picolinamide and nicotinamide are indistinguishable to the inverse task, but their forward-predicted ^{13}C spectra are not: the true isomer matches the observed spectrum at a chamfer of 0.42 ppm against 1.30 ppm for the alternative. This is the LLM analog of the DP4 / NMR-crystallography logic, with a forward language model in place of the quantum chemistry (§5.1).

ppm (quartiles 0.84 and 1.78), and **82% of such pairs are predicted closer together than the predictor's own ~2 ppm error**. Non-isomeric pairs separate better but not dramatically (median 1.90 ppm, 53% inside the error). So the honest form of the premise is not that regioisomers are predicted far apart; it is that they are predicted *slightly* apart, usually within the noise, and that the ranking nevertheless recovers the right one 89% of the time when it is present (§5.2) — a real effect on a thin margin, confirmed against a derangement null at p=0.001 (§5.5).

This single measurement is the quantitative root of three limits the rest of §5 reports separately: precision falling 84%→72% as more near-degenerate isomers enter the pool (§5.3), the 74% derangement chance floor (§5.5), and the near-degenerate ceiling that neither the HOSE lookup nor the GNN escapes (§5.4). All three are the same fact seen from different angles — the predictor's resolution and the isomers' spacing are of the same order, so a sharper predictor, not a better search, is what would move them.

That is a concrete, checkable prescription rather than a gesture, because predictors of the required sharpness already exist. Message-passing ensembles trained on large assigned-shift corpora reach ~1 ppm on ^{13}C,[31] and coupling a graph network to DFT-computed shielding tensors reaches 0.94 ppm[32] — roughly half the 1.21 ppm median spacing we measure between competing candidates, where our LLM predictor sits at or above it. A community benchmark now exists to compare such models on common ground,[33] alongside curated experimental shift sets for calibrating them.[34] Dropping one into the verifier slot is the single change most likely to move §5.2's 89%, and §5.4 takes a first step in exactly that direction with a model we train ourselves.

Conceptually this is **analogous to NMR-crystallography logic, with an LLM in place of the quantum chemistry** — an analogy, not an equivalence: where DP4/DP4+ rank candidates by GIAO-DFT shifts[16,17] and NMR crystallography adjudicates polymorphs and connectivity by GIPAW-computed shifts[18,19], we rank by the shifts a forward LLM predicts. The shared principle is *verification by forward prediction*; what we do **not** inherit is DFT's calibrated error model. The trade is deliberate — we forgo the ≈1–2 ppm accuracy and the rigorous, probabilistic error model of DFT (the very thing that lets DP4 emit a posterior probability) for a predictor that runs in seconds at zero setup, and we quantify below exactly how far that cheaper, uncalibrated verifier carries the inverse problem.

5.2 Result

We ran the verifier over **every compound in the benchmark**: 373 deduplicated candidate structures across all 194 targets, all of them forward-predicted blind, in shuffled and anonymised batches, by 23 independent agents. The first column below is the original 60-compound arm (v3 + v2-control), on which §5.3–§5.5 build; the second is the full benchmark, where the recall-conditional claim actually lives.

Table 6. Forward-verification decomposition, on the original arm and on the whole benchmark.

	60-compound arm	**full benchmark (n=194)**
generation recall	19/60 (31%)	**65/194 (34%)**
top-1, solver self-ranking	14/60 (23%)	55/194 (28%)
top-1, **forward-verified re-ranking**	16/60 (26%)	**58/194 (30%)**

	60-compound arm	**full benchmark (n=194)**
conditional on recall — self-ranking	14/19 (73%)	55/65 (**85%**)
conditional on recall — **forward-verification**	16/19 (**84%**)	58/65 (**89%**)
...single-candidate compounds within that set	6/19	28/65
multi-candidate only — self-ranking	8/13 (62%)	27/37 (73%)
multi-candidate only — **forward-verification**	10/13 (77%)	**30/37 (81%)**

Two facts make the full-benchmark column trustworthy before its content is read. Its self-ranking row is derived, by an independent script over the released candidate files, from the same solver output as §4, and it lands on Table 2 exactly: 55/194 = **28.4%** top-1 and 65/194 = **33.5%** recall, and the same in both strata (simple 48.0% / 54.1%, complex 8.3% / 12.5%) — all six headline numbers re-derived, digit for digit, through a separate code path. And of the 373 candidates, **373 were forward-predicted**: every candidate the verifier could rank was rankable, so nothing here is a lower bound. (The same is now true of §5.3, whose coverage gap we also closed.)

The decomposition is the finding. **When the true structure is among the candidates, forward verification selects it in 58 of 65 cases (89%)** — high in absolute terms and ~15 points above the 74% derangement chance floor of §5.5 (given how few and near-degenerate the recall-positive candidate sets are). Extending the arm from 19 recall-positive compounds to 65 did not soften the claim; it firmed it, and it moved the §5.5 permutation control from marginal (one-sided p=0.019) to unambiguous (**p=0.001**).

One composition detail must be stated, because it inflates every ranker equally and §5.5 excludes exactly these cases from its own analysis. Of the 65 recall-positive compounds, **28 had a single candidate**, so the verifier faced no choice and any ranker — ours, the solver's, a coin — scores them by construction. On the **37 compounds where a choice actually existed**, forward-verification selects the true structure in **30/37 (81%)** and the solver's own self-ranking in **27/37 (73%)**. The asymmetry the paper rests on is unaffected: 34% generation recall against 81% verification precision is the same wall. We report both denominators throughout and treat the 37-compound figure as the one that measures verification.

Pooling the two arms is licensed rather than assumed. They agree on every quantity the pooled column reports — verification precision conditional on recall (42/46 vs 16/19, Fisher exact p=0.41), the same restricted to multi-candidate compounds (20/24 vs 10/13, p=0.68), self-ranking on that subset (19/24 vs 8/13, p=0.28) — and even on the composition detail above, the single-candidate fraction (22/46 vs 6/19, p=0.28). Table 6 keeps them in separate columns anyway, so a reader who prefers the original arm can read it unpooled.

Either way the margin over self-ranking remains a small, unresolved difference: seven compounds gained and four lost, McNemar exact **p=0.55**. Quadrupling the conditional-on-recall set did not turn it significant, and we do not claim it. The load-bearing claim is that verification precision is high **in absolute terms** while recall binds — not that forward-verification beats self-ranking. The overall top-1 moves only 28%→30% because the binding constraint is **generation recall**: the true structure was never proposed for 129 of 194 compounds, which no re-ranking can repair.

The difficulty split sharpens where the verifier earns its keep and where it does not. On *simple* targets the verifier converts recall almost completely (50/53, **94%**, against self-ranking's 47/53);

on *complex* targets it converts 8/12 (67%) and is **exactly tied** with self-ranking, gaining two compounds and losing two. The ~2 ppm forward predictor resolves the regiochemistry of tractable targets and cannot resolve the near-degenerate alternatives that hard targets generate — the same precision ceiling §5.3 hits from the other direction, here visible as a clean stratification rather than an aggregate.

LLM elucidation therefore factorises into two near-independent levers: the **verifier is already strong (89%)**; the **generator (34% recall) is the wall** (Fig. 1). Table 6 regenerates from the released artifacts via `scripts/forward_verify_all.py`.

5.3 Generate-wide: testing the recipe

The decomposition implies a recipe — *generate wide, verify by forward prediction* — a chemistry analog of self-consistency sampling, where many independent generations are pooled and the answer is chosen by agreement rather than from a single pass.[35] We tested it directly: ten independent solver agents proposed up to **six regiochemistry-aware candidates per compound**, pooled with the originals, and the 65 new candidates were forward-predicted and re-ranked as before. The measured result:

Table 7. Generate-wide vs original, both on the 60-compound arm. The "original" column is Table 6's first column, not its full-benchmark one; wide generation was run on this arm only, so the comparison is held to it.

	original (60-compound arm)	generate-wide
recall (true structure among candidates)	31%	**41%**
forward-verified top-1	26%	**30%**
verification precision (conditional on recall)	84%	72%

Wide generation lifts recall +10 points (31%→41%, i.e. 19/60→25/60) and exact top-1 +7 points over the self-ranking baseline (23%→30%, 14/60→18/60) — equivalently +4 over the original forward-verified top-1 (26%→30%, 16/60→18/60, Table 7) — on the same 60 compounds (Fig. 6), with no training.

The arm originally forward-predicted only **65 of the 217** distinct new candidates, which made its top-1 a lower bound rather than a measurement: an unpredicted candidate is assigned an infinite match distance and can never be selected, so recall could count every candidate while top-1 benefited only from the predicted subset. **We have since predicted all 217, and not one number moves** — recall 41%, forward-verified top-1 30%, precision 72%, identical to three significant figures. The bound was tight.

What the added coverage does buy is a direct view of the mechanism. On **18 of the 60 compounds the verifier abandons its previous pick for a newly-selectable candidate** — so the extra candidates genuinely compete, and often win on chamfer distance — yet in **not one** of those 18 does the outcome change: every switch is from one wrong structure to another. The wide pass supplies more near-degenerate alternatives, the ~2 ppm predictor prefers some of them, and it cannot tell them from the truth. That is the precision ceiling of the paragraph below observed directly rather than inferred, and it is why recall — not verification, and not prediction coverage — remains the binding constraint. Table 7 and this coverage analysis regenerate from the released artifacts via `scripts/score_generate_wide.py` and `scripts/forward_verify_gw.py`

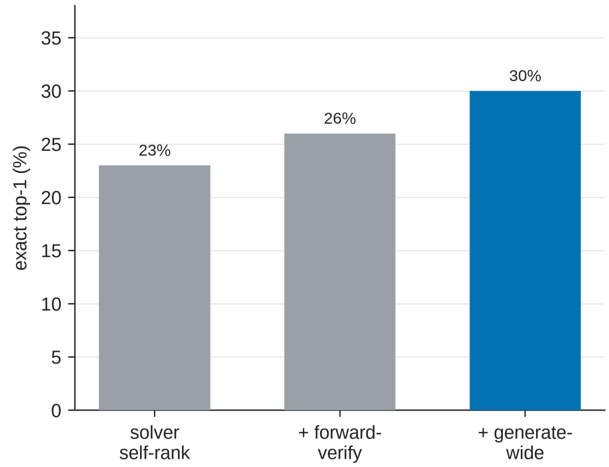

Fig. 6: The forward-verification inference ladder on the 60-compound arm: solver self-ranking → + forward-verification → + generate-wide, at 23% / 26% / 30% top-1. Each rung is a training-free change to inference alone. The steps are directional rather than individually resolved at this sample size — the paired tests are in §5.3.

(`data/fverify_gw/results.txt`). **These top-1 differences are directional, not statistically resolved at n=60:** the 14/60→18/60 improvement is a four-compound difference, but the stages are **not** nested: going from self-ranking to generate-wide gains seven compounds and loses three, so the paired test is McNemar exact **p=0.34** (b=3, c=7). The intermediate steps are weaker still — self-rank→forward-verify p=0.63 (b=1, c=3) and forward-verify→generate-wide p=0.69 (b=2, c=4). Discordant counts are computed from the released per-compound outcomes by `scripts/ladder_significance.py`. The recall gain is the better-supported effect. We therefore read the ladder as a demonstration that the *mechanism* works and that recall is the movable factor, not as a precise measurement of the size of the top-1 gain. But it does **not** reach the ~50% a naïve extrapolation would predict, for two instructive reasons: **(i) recall plateaus at 41%** — and the compounds it plateaus on are characterised by size and ring count rather than by exotic elements. Comparing the 65 recall-positive against the 129 recall-negative compounds: median heavy atoms 16 vs 23, median ring count 1 vs 3, and **≥4 rings in 3.1% of recalled compounds against 38.0% of missed ones**; nitrogen density separates them too (≥4 N: 4.6% vs 17.1%). Selenium heterocycles and the like are memorable but marginal — Se appears in 2.3% of misses and none of the hits, and S, F, Cl and P show no separation whatsoever. Polycyclic, nitrogen-rich, large targets resist even six regiochemical variants; that is the pattern, and it is a mundane one (`scripts/analyze_misses.py`). And **(ii) verification precision falls 84%→72%** as more near-degenerate regioisomers enter the pool and the ~2-ppm forward predictor can no longer separate them. This recall/precision tension is the honest ceiling of the training-free approach: the recall-bound diagnosis stands, and closing the gap further requires either sharper verification or 2D-NMR constraints, not merely more candidates.

5.4 Non-LLM verifiers: a deterministic lookup and a learned model

§5.3 suggests an obvious fix: swap the LLM forward-predictor for a non-LLM ^{13}C predictor with a rigorous error model. We test two, both trained on the **same nmrshiftdb2 dump**[36] and both dropped into the verifier slot on the **same §5.2 candidate sets** — the 60-compound arm and the full benchmark — so that only the predictor changes.

The first is the canonical choice: a **HOSE-code[37]-style lookup** (RDKit radial-environment bins, spheres r=4→1 with a hybridisation prior fallback; 31,000 molecules, 332,595 assigned carbons; held-out MAE 3.23 ppm, median 1.73). The second is a small **message-passing GNN** (4 layers, per-carbon ^{13}C regression) trained on the identical dump, which reaches held-out MAE 1.70 ppm (median 1.02) — roughly twice as sharp, and an independently competent predictor. It is not a state-of-the-art one, and we do not present it as such: purpose-built ^{13}C models reach ~1 ppm from message-passing ensembles[31] and 0.94 ppm when a graph network is coupled to DFT shielding tensors,[32] on benchmarks now standardised for the comparison.[33] Ours is deliberately modest, because the question here is *whether the predictor slot is where the leverage sits*, and answering that needs a predictor that differs from the lookup in method while holding data and evaluation fixed — not the best model available.

Table 8. Verifier comparison, conditional on recall, on identical candidate sets. Only the ^{13}C predictor changes between rows.

verifier	60-compound arm (n=19)	full benchmark (n=65)	held-out ^{13}C MAE
solver self-ranking	14/19 (73%)	55/65 (85%)	—
deterministic HOSE *lookup*	14/19 (73%)	55/65 (85%)	3.23 ppm
learned GNN (same data)	16/19 (84%)	**59/65 (91%)**	**1.70 ppm**
LLM forward-verifier (§5.2)	16/19 (84%)	58/65 (89%)	—

The lookup does not help, and quadrupling the evaluation set did not rescue it: it lands on exactly the solver's own score at both sample sizes. That tie is not agreement — at n=65 the lookup **gains seven compounds and loses seven** against self-ranking (McNemar b=c=7, p=1.00). It reshuffles the ranking energetically and carries no net discriminative signal. Its diagnosis is coverage — of the 6,360 candidate carbons, only **2%** match a training environment at the most specific sphere (r=4) and **71%** resolve only at coarse spheres (r≤2) or fall through to the hybridisation prior, because the benchmark's exotic chemistry (selenium heterocycles, polyaryl ketones, large polyamines) is under-represented in nmrshiftdb2. A coarse environment cannot separate the regioisomers the verifier exists to separate. (The same diagnostic on the 60-compound arm gives 2% and 70% over its 2,244 carbons; both regenerate via `scripts/hose_predict.py coverage`.)

The learned model, on the same data, matches and slightly exceeds the LLM verifier — 59/65 against 58/65 (Fig. S6). Its margins over the two baselines survive the larger set and grow more credible without reaching significance: against the lookup, seven compounds gained and three lost (McNemar exact p=0.34, from p=0.63 at n=19); against self-ranking, five gained and one lost (p=0.22, from p=0.50). We therefore keep the same reading the smaller set supported, now on four times the evidence and with the direction unchanged in every pair: the two rows are *suggestive and directional*, consistent with the lookup's failure being substantially **method** — an inability to generalise across novel environments — rather than coverage alone, and with a cheap learned predictor being sufficient to match the LLM without compound-specific DFT accuracy or 2D-NMR. They do not establish it. What none of these reach is the near-degenerate-regioisomer precision ceiling (§5.3/§5.5), where DFT-level accuracy or orthogonal 2D-NMR constraints remain the genuine fix.

The GNN and the LLM land within one compound of each other while **disagreeing on nine** (b=5, c=4, p=1.00) — two verifiers built on unrelated evidence, concurring in aggregate rather than one echoing the other. The result is **generalisation, not memorisation**, and we can now say so on the whole candidate pool rather than a sixth of it. *Exact* overlap with the entire nmrshiftdb2 database is **2 of 364** distinct candidate structures by InChIKey-14 — both of them *wrong* candidates on *recall-negative* compounds that never enter the conditional analysis — and **no benchmark answer appears in the database at all** (0/373; the 60-compound arm is 0/126, as previously reported). *Analog* overlap, which exact matching would miss, is likewise absent: the median Morgan(2, 2048) Tanimoto to the nearest training molecule is **0.44** and only three candidates exceed 0.80. The load-bearing number is the last one: over the **65 true structures the verifier actually has to identify**, the nearest training analog has median Tanimoto **0.50** and maximum **0.81** — not one of the compounds the conditional analysis scores has a near-duplicate in the training set. A Y-randomisation control (1,000 derangements) places the real result above the 97.5th percentile of the chance distribution (n=19: real 84% vs mean 58.6%, 95% range 42.1–73.7%, one-sided $p<0.05$). Like §5.6, the learned verifier is a **trained complement**, reported outside the training-free protocol. Table 8 and both leakage checks regenerate from the released artifacts (`scripts/verifier_table.py`, `scripts/verifier_leakage.py`, `data/fverify/verifier_table_results.txt`, `data/nmrshiftdb/gnn_c13.pt`; `docs/VERIFIER_PROBE.md`).

5.5 Negative control

Permutation negative control (Y-randomisation analog). A re-ranker earns the interpretation that it exploits genuine predicted-vs-observed ^{13}C agreement only if that agreement is destroyed when the pairing is broken. Borrowing the Y-randomisation discipline of QSAR validation, we re-paired — as a *derangement*, so no compound keeps its own spectrum — which observed ^{13}C spectrum each candidate set is scored against, and re-ran the verifier 1,000 times. On the full benchmark, conditional-on-recall precision falls from the true **89.2% (58/65)** to a permuted mean of **73.8%** (95% range 66.2–81.5%; one-sided empirical p=0.001, two-sided p=0.002) — the verifier acts on real spectral agreement, not a candidate-list artefact. The same control on the 60-compound arm alone gives 84.2% against a permuted 66.4% (one-sided p=0.019): same effect, four times the evidence. The honest caveat is the height of the chance floor: because recall-positive compounds carry few and near-identical (regioisomeric) candidates, even a random pairing lands on the correct structure ~74% of the time, so the verifier's genuine margin over chance is ~**15 points**, not the full 89% — real and significant, but to be read against this high baseline.

Confidence calibration (a negative result). We also asked whether the chamfer match-distance doubles as a confidence signal that would support selective prediction. It does not. Ranking the 138 multi-candidate compounds by their chamfer margin (the gap between the best and second-best predicted-vs-observed distance) and answering only the most-confident fraction leaves top-1 essentially flat and non-monotonic with coverage (22% at full coverage, 24% at 75%, 28% at 50%, 24% at 25% — neither rising nor falling reliably as the threshold tightens). Compounds with a single proposed candidate have no margin at all and must be excluded; an earlier analysis that retained them produced a spurious "improvement" that was entirely an artefact of those trivial cases. We therefore do **not** claim the verifier distance as a calibrated abstention gauge — a sharper, DFT/2D-NMR-grounded confidence estimate (§5.4) remains the route to reliable selective prediction.

Both controls in this section regenerate via `scripts/verifier_diagnostics.py --all` (the full benchmark, as reported here); omitting `--all` reproduces the 60-compound arm's figures instead.

5.6 Is the recall wall task-intrinsic? A trained-generator probe

The ceiling above is a property of *training-free LLM elicitation*; it leaves open whether the ~41% recall plateau is intrinsic to 1D-data elucidation or specific to how LLMs are elicited. We test this with a complementary probe — a small (~16M-parameter) $^{1}H/^{13}C \rightarrow$SMILES transformer (simulated-spectra pretraining, then fine-tuned on IRexp; ensemble of four), held **deliberately separate from the training-free protocol**. Its candidates are filtered to the given molecular formula, pooled with Claude's, and re-ranked by the same verifiers.

The generator supplies candidates enumeration cannot. On the 194-compound benchmark the true structure enters the candidate pool for **54.1%** of compounds (versus 41.8% for scaffold enumeration and 33.5% for Claude alone). Crucially, where enumeration's near-degenerate regioisomers *collapse* the deterministic HOSE verifier (top-1 28.4%→16.0%, the §5.3 precision-loss mechanism), the generator's formula-correct, ^{13}C-separable candidates **convert**: HOSE top-1 rises **28.4%→35.1%** (McNemar exact p=0.015; +6.7 points, 95% CI [+2.1, +11.9]; Fig. S5). With the stronger LLM forward-prediction verifier on the 60 forward-verify compounds, recall rises 41%→**56%** (34/60) and forward-verified top-1 reaches **46%** (28/60), at 82% precision conditional on recall (28/34).

Those last two figures are a **re-run, not a reproduction**, and the distinction matters. An earlier pass reported 41% top-1 at 73% precision, but its forward-prediction outputs were never deposited, so that number was unverifiable and we do not stand behind it. We therefore re-ran the arm from scratch under the identical blind protocol — the 75 outstanding candidates forward-predicted by agents that saw the anonymised SMILES and nothing else — and **released every prediction** (`data/fverify_gen/raw/`), so `scripts/forward_verify_gen.py score` now regenerates all three numbers from the bundle with no missing predictions. The re-run lands *above* the figure it replaces (46% vs 41% top-1, 82% vs 73% precision); we report the reproducible one and flag that it was collected later than the June solver runs (`docs/MODELS.md`). The recall arm and the deterministic-HOSE re-rank were reproducible throughout and are unchanged.

Two controls guard against memorisation, both reproducible from the released bundle (`contrib/generator_probe`): (i) the fine-tuning split removes every benchmark InChIKey-14 (train∩benchmark = 0, val∩benchmark = 0), and **none of the 40 newly-recovered compounds appear in either training stage** (0/40 in simulated pretraining, 0/40 in fine-tuning); (ii) the simulated-pretrained model alone recovers **0/248** benchmark structures zero-shot, rising to 25% only after IRexp fine-tuning. The IRexp data — not the architecture — is the active ingredient, and the recall ceiling is therefore **elicitation-specific, not task-intrinsic**. We report this as a probe, not part of the headline training-free protocol; it answers whether the wall is breakable, not whether to abandon training-free methods.

Two published systems make this point far more strongly than our 16M-parameter probe can, and we would rather lean on them than on ourselves. NMR-Solver reaches 52.9% top-1 on experimental literature $^{1}H/^{13}C$ with the formula supplied, using a purpose-trained shift predictor inside the same generate-and-verify loop[3]; and a purpose-trained IR transformer reaches 63.8% top-1 on experimental NIST spectra in the 6–13-heavy-atom range[15]. Whatever bounds a training-free LLM at 28–30%, it is plainly not the task. Our probe adds one thing those results do not: it isolates the *data* as the active ingredient (0→25% with IRexp fine-tuning, 0/248 without), which is the claim IRexp itself has to earn. (The released public split `irexp_release/train` does *not* hold out the benchmark — it overlaps it by 117/200 InChIKey-14 — so downstream users must de-leak as in `build_exp_manifest.py`; see Data and code availability.)

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6. Discussion

The frontier LLMs we tested (the Claude family) are neither "solving structure elucidation" on realistic 1D data nor failing at it. They are reliable **scaffold-level** elucidators (best-candidate Tanimoto ≥ 0.45 for 56% of compounds, 73% of simple ones; mean best Tanimoto 0.59) and good **verifiers** (89% conditional on recall). Exact top-1 is throttled by candidate recall and by the regiochemical underdetermination intrinsic to 1D NMR. The contribution of this paper is therefore best read as a **diagnosis with a bounded, training-free improvement attached**, not as a method that solves the task. The diagnostic result is the durable take-away, and it held under every perturbation we were able to apply — four Claude models, a second chemical domain, four different verifiers: the wall is **generation recall**, not verification. Whether it holds outside the Claude family is the open question, not a settled one (§7). The accompanying improvement is real but deliberately modest — forward verification alone moves top-1 from 28% to 30% across the whole benchmark (§5.2), and adding wide generation takes 23% to 30% on the 60-compound arm where that was run; we show by direct measurement (§5.3) that this stays *below its own ceiling* (recall plateaus at 41%; verification precision falls to 72% as near-degenerate regioisomers accumulate), while the match distance fails as a confidence gauge for selective prediction (§5.5, a reported null result). This reframes the engineering problem. Rather than training a bespoke spectra→structure model — a target the frontier already meets at the scaffold level and that ages out each model generation — the durable levers are (i) **open, hard, honestly scored benchmarks** that expose specific failure modes (here, regiochemistry and recall), and (ii) **inference-time scaffolding** (decoupled solving, generate-and-verify) that needs no training and improves with each model. IRexp's IR modality and 2D-NMR-ready provenance position it for the obvious next step: supplying the HMBC/COSY/NOESY constraints that would attack regiochemistry at the source. Our trained-generator probe (§5.6) sharpens the point: the recall wall is *elicitation-specific, not task-intrinsic* — a small generator fine-tuned on IRexp lifts it (recall 41→56%) — and because that gain comes entirely from the released data (0→25% with fine-tuning, 0/248 without), it is the **open dataset**, not a bespoke architecture, that does the work when training does help.

Protocol is a lever of the same order as capability. How a problem is posed and verified — bounded, reset contexts with tool access; generate-and-verify — moves accuracy enough that it must be reported alongside any capability claim. We do *not* claim it dominates capability: on the fixed 24-compound subset the model axis spans 0% to 45% top-1 (Table 3), a wider range than the 5%→15% protocol effect in §4.3, and the two are measured on different sets. The point is that a number quoted without its protocol is uninterpretable, which echoes a recurring lesson across cheminformatics: careful pipeline and protocol design competes with the fashionable component. The same pattern appears in molecular property prediction, where a recent systematic benchmark reports that learned, pretrained molecular embeddings rarely outperform classical ECFP fingerprints once evaluation is controlled[38] — the modelling fashion underperforms the well-engineered baseline. We read our within-compound control (§4.3) and forward-verification recipe (§5) in the same light: the durable gains in LLM elucidation come from honest benchmarking and inference-time scaffolding rather than from waiting on the next, larger model. We draw this parallel narrowly and claim no formal equivalence between the two settings.

That the bottleneck reproduces inside a single application domain (§4.5: 26% top-1 on battery-electrolyte chemistry, n=46, against 28% overall) is consistent with it being structural rather than incidental — and it makes the forward-verification recipe directly relevant to the magnetic-resonance workflows (electrolyte-decomposition assignment, NMR crystallography) where computing a candidate's spectrum to confirm it is already standard practice.

For practitioners, two operational findings transfer immediately: solve each problem in bounded, frequently-reset contexts with tool access (5%→15%, 1/20→3/20, directional at n=20, McNemar p≥0.25, §4.3), and use forward-predicted-vs-observed ^{13}C agreement to *re-rank* candidates, not to decide whether to trust the winner. §5.5 is explicit on this: match distance is a strong relative re-ranker but failed as an absolute confidence gauge in our selective-prediction test, so it does not support abstention thresholds.

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7. Limitations

Several limitations of earlier drafts are now resolved with controlled experiments; we state what remains plainly.

Resolved. **Extraction noise:** an RDKit self-consistency audit (^{13}C peak count vs symmetry-unique carbons, formula, SELFIES round-trip) finds **57/60 ground truths spectrally clean**, and the metrics are unchanged on that clean subset (self-ranking top-1 24% vs 23%, recall 33% vs 31% on the n=60 forward-verify set) — the conclusions are not driven by scraping artefacts. **Verifier sample size:** forward verification now runs over **all 194 compounds**, not a 60-compound subset, so the recall-conditional claim rests on n=65 rather than n=19 and the §5.5 permutation control tightens from one-sided p=0.019 to p=0.001 (§5.2). **Verifier precision / abstention:** the generate-wide experiment (§5.3) quantifies the recall/precision tension directly — verification precision is 72–89% conditional on recall and degrades as near-degenerate regioisomers accumulate, so forward-match distance is a strong *re-ranker* but a soft *confidence* gauge. We further tested non-LLM verifiers (§5.4), on the same extended set: a nmrshiftdb2-trained HOSE-code *lookup* does **not** beat the LLM verifier (85% vs 89% conditional at n=65) and ties self-ranking by exchanging seven compounds for seven, while a *learned* GNN on the same data reaches 91% — directional in every pairwise comparison but still not statistically resolved at n=65 (p=0.34 vs the lookup, p=0.22 vs self-ranking), so it is suggestive that the deterministic failure was method rather than coverage alone, not a demonstration of it. What remains beyond all of them is the near-degenerate-regiochemistry precision ceiling, where DFT-level accuracy or 2D-NMR constraints are still the genuine fix. **Projection:** §5.2's extrapolation is replaced by the **measured** §5.3 result (top-1 30%, recall 41%) — and is honestly below the optimistic estimate.

Independence checks. Scoring throughout is mechanical RDKit (not LLM-judged), and all solver/verifier runs were transcript-audited *at generation time* for zero web and zero ground-truth access. That audit is the one part of the pipeline a reader **cannot re-verify from the release**: the committed artefacts are the parsed per-compound predictions, and the transcripts themselves are not deposited (they are available on request). We state this rather than let "transcript-audited" imply an independently checkable artefact, and note that the *outcome* of closed-book solving is separately testable from what is released — the formula-only control (§4.6) removes the spectra and accuracy collapses, which no amount of web access would produce. To confirm that the forward-verifier's measured advantage reflects real predicted-vs-observed spectral agreement rather than leakage or a candidate-list artefact, we ran a permutation negative control (Y-randomisation analog, §5.5): re-pairing — as a derangement — which observed ^{13}C spectrum each candidate set is scored against, over 1,000 permutations, collapses conditional-on-recall precision from 89.2% to a chance mean of 73.8% (two-sided p=0.002 on the full benchmark; p=0.038 on the 60-compound arm alone) — the signal weakens to the chance floor under label-shuffle, as a correctly isolated blind evaluation requires. A second Claude model (Sonnet) re-solved a 12-compound subset under the identical generate-wide protocol and was comparably recall-bound (recall 33% vs Opus 41% on those compounds), consistent with the recall bound being a property of the task rather than of one model. As this is within the Claude family, it speaks to model-instance robustness, not cross-vendor generality.

Remaining. **(i) Pretraining contamination is bounded, not excluded.** Every benchmark compound was mined from open-access literature, so a frontier model may have encountered it in training. The formula-only control (§4.6) bounds how much of the headline number pure recall can explain — masking the spectra drops top-1 from 23% to 5%, perfectly nested, McNemar exact p=0.001. §4.6 examines the three formula-only successes one by one rather than explaining them away collectively: only one (an unlisted silyl aryl sulfonate) has a genuinely determining composition, while the other two — *N*-tosyl-leucine and the natural product [6]-paradol — are catalogued compounds for which recall is a live explanation. That is the control working as intended, and it leaves the bound unchanged. The second control (§4.6) resolves the source publication year for all 194 compounds and finds accuracy flat in it ($r=-0.007$), with the size-adjusted older-versus-newer difference bounded at -5.1 points, 95% CI [-17.2, $+7.0$].

Two things these controls do not do. Neither is randomised: publication year is observational, and the formula-only arm cannot separate memorisation from inference on a near-determining formula. And we do not anchor the recency test to a *known* training cutoff — the subscription harness does not disclose one (§8), so we test recency as a continuous proxy rather than partitioning at a cutoff date. A replication restricted to compounds published after a disclosed cutoff would settle what these bound, and remains open.

(ii) Human audit — prepared but not yet run. This covers two things we have not validated by hand: the elucidation and forward-prediction outputs, and the *recall* side of dataset extraction (whether the parser found every IR string in every source paper). Transcription fidelity of the records we do hold is measured and high (§2.3), but record-level recall is not, and only reading papers can settle it. Solver and verifier are both LLMs, so the one validation we cannot perform ourselves is an expert-chemist review of a sample of elucidations and forward predictions. We have therefore **built and frozen** a blinded, pre-registered audit package — a difficulty-stratified 30-compound sample (9 recall-positive), rendered candidate structures, a separate withheld answer key, and a mechanical scoring sheet — released at `data/audit/` with its protocol at `docs/EXPERT_AUDIT_PROTOCOL.md` (regenerable via `scripts/make_audit_sample.py`), designed to test the two load-bearing claims (what a miss actually is, §4; forward verification is a trustworthy re-ranker, §5). **We have not yet run the panel**; until expert results are in, those claims should be read as machine-validated (RDKit InChIKey) but not yet human-validated.

The panel's first question has since been narrowed by measurement rather than left to judgement. §4 now reports the composition of all 139 misses mechanically — 76.6% constitutional isomers, 22.6% scaffold-preserving positional errors — which replaces the impressionistic claim the audit was built to check and, in doing so, corrected it: the failures are predominantly *wrong connectivity at the right composition*, and strict regiochemistry is a minority of them. What remains for a chemist is the part no fingerprint settles: whether a formula-correct, scaffold-wrong candidate is a *chemically reasonable* reading of the spectra or an implausible one. That is a narrower and better-posed question than the one we started with.

(iii) Single vendor and an underpowered cross-model comparison. Every number comes from one vendor's models. The headline (28.4% top-1) is a single frontier model (Claude Opus), and the cross-model evidence (§4.4) is four **Claude-family** models on a shared **24-compound** subset. As §4.4 sets out, that comparison is underpowered: the ranking is robust (strictly nested, weakest floored at 0%) but no adjacent gap is established at n=24. Quantitative claims should therefore be read as holding **for the Claude family**, and a true cross-**vendor** sweep (GPT-, Gemini-class, open-weight models) — the test of whether the pattern is a property of the task rather than one lineage — was forgone because it needs paid API access incompatible with our zero-cost protocol; we flag it as the most important external-validity experiment left open.

(iv) Domain-subset scope: the battery-electrolyte case study (§4.5) comprises *literature compounds bearing electrolyte-relevant functional chemistry* drawn from the open corpus — not operando or in-cell decomposition spectra — so it demonstrates functional-class transfer of the elucidation bottleneck, not direct assignment of authentic interphase/degradation products, which would require a dedicated operando set.

(v) Constitution-only scoring: correctness is judged on InChIKey connectivity, so a correct-constitution / wrong-stereochemistry prediction is counted correct. Scoring the full InChIKey instead gives 21.1% top-1 (§3), 7.3 points lower, so the headline should be read as an upper bound on full-stereochemistry accuracy. We keep constitution as the headline because 1D NMR/IR rarely determines absolute configuration — the 10.3% (20/194) of targets with a defined stereocentre would be penalised for information the prompt never carried — but a stereochemistry-sensitive benchmark would need 2D/chiroptical data.

(vi) Single-sample scoring: each headline compound is scored from one solver prediction set (one decoupled run), so reported top-1/recall carry no run-to-run (LLM-sampling) variance estimate — the bootstrap CIs reflect compound sampling only. The generate-wide experiment (§5.3) pools ten independent generation passes (recall 31%→41%), indicating generator stochasticity that single-pass scoring understates.

(vii) Chemical-space coverage: the benchmark is drawn from open-access organic methodology and total-synthesis literature, covering small-to-medium drug-like and synthetic-organic space; it is **not** representative of organometallic/coordination compounds, large biomolecules (peptides, oligonucleotides, oligosaccharides), or stereochemistry-heavy targets, and our results should not be extrapolated to them.

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8. Methods

Mining and resolution. PMC-OA full text was fetched from `s3://pmc-oa-opendata`; records were extracted with a deterministic parser and resolved with OPSIN, RDKit, and SELFIES, with an optional cached PubChem fallback. Durable content-keyed caches make resolution additive and restart-safe.

Benchmark and agents. Problems were sampled from `irexp_resolved`, stratified by RDKit ring analysis, with cross-round de-duplication by InChIKey. The battery-electrolyte subset (§4.5) was drawn from the same corpus by SMARTS substructure filters for six electrolyte functional classes (carbonate, sulfonyl/sulfonate, nitrile, sp^3-C–F, phosphoryl, glyme/oligoether), balanced to eight compounds per class (48 curated; 46 scored after two yielded no parseable candidate), excluding every compound used elsewhere, and J-enriched and spectrally validated identically to the main rounds. Solver and forward-prediction agents were independent Claude-Opus sub-agents invoked under a consumer subscription; agents were instructed closed-book and audited via automated transcript search for zero web/answer access. Scoring used RDKit InChIKey-connectivity matching and Morgan(2, 2048) Tanimoto; forward-verification used a symmetric chamfer distance over ^{13}C peak sets. The core protocol trains no model and uses no paid API; the two trained probes — the §5.6 generator and the §5.4 learned ^{13}C verifier — are the only separately-trained exceptions, each reported as a complement and fenced from the headline results.

Models and versions. All experiments used Anthropic Claude models via the consumer subscription (claude.ai). The §4.4 comparison spanned, in capability order, Claude Haiku, Claude Sonnet, Claude Opus, and Claude Fable 5; the headline benchmark (§4.1) and forward-verification (§5) used Claude Opus. `docs/MODELS.md` records, for each experiment, the model used, its data directory, the collection date, the harness and tool access, and the scoring code path. Model invocations fall into three dated windows, listed separately there because no single window covers them. Every candidate structure behind the **headline** results — §4.1, §4.3–§4.5, and the candidate pools all of §5 re-ranks — was generated between **2026-06-09 and 2026-06-11** (UTC). The formula-only contamination control (§4.6) is a deliberate exception: it re-solves the same 60 compounds with the spectra masked on **2026-07-28**, generating its own candidates, because a masked-input control is only meaningful as a fresh run; it touches Table 5 alone. Three **2026-08-07** collections forward-predict ^{13}C for candidates the June run had already produced (the §5.2 extension to all 194 compounds, the §5.3 coverage-gap closure, and the §5.6 re-run); none introduces a new candidate structure, and no recall number in the paper changes. One *verified* number does: the §5.6 re-run supersedes a top-1 whose original forward-prediction outputs were never deposited and are lost, and it does not agree with the figure it replaces (46% against 41%). §5.6 reports the reproducible one and says plainly that it is a re-run rather than a reproduction. Decoding parameters are not exposed by the subscription harness and were neither set nor recorded, so re-running reproduces the protocol distributionally rather than exactly.

No model snapshot can be reported, and this is a property of the protocol rather than an omission. The consumer harness exposes no checkpoint identifier to the caller, announces no build changes, and records nothing about which build served a given request; no work after the fact recovers that. Two consequences follow. `Claude Opus 4.8` is evidenced only for the 2026-06-09 pilot, so **a mid-window build change cannot be excluded** and we do not claim the same build served the main round two days later. And a reader repeating these experiments will be served whatever build is then current, so distributional agreement is the most that can be expected — exact agreement is not a meaningful target here and we do not offer it as one. What stands in place of a pin is everything that *is* fixed: the dated collection windows above, the frozen per-compound outputs, and mechanical scorers that regenerate every number from them. That is the boundary of a zero-cost subscription protocol: it buys exact reproducibility of scoring and analysis, not of inference (`docs/MODELS.md` §6).

Reproducibility. Every round is frozen: questions, ground-truth answers, per-agent raw outputs, predictions, and scorer outputs are released, and the sampler, scorer, and forward-verification harness are scripted end-to-end.

Supporting Information figures

These are supplied as a separate Electronic Supplementary Information document (`docs/paper_esi.pdf`, built by `scripts/build_pdf.py`), as RSC requires; they are listed here for reference.

- **Fig. S1** (`docs/figures/fig0_overview.png`) — study design: open multimodal data (IRexp) → blind, complexity-stratified benchmark → decoupled blind solving → forward-verification re-ranking; training-free core pipeline.
- **Fig. S2** (`docs/figures/fig4_dataset.png`) — IRexp composition: IR records → NMR-paired → structure-linked → full IR+^{1}H+^{13}C+structure quadruples.
- **Fig. S3** (`docs/figures/fig2_size.png`) — accuracy vs molecular size (heavy-atom bucket); the monotonic 60%→7% top-1 gradient.
- **Fig. S4** (`docs/figures/fig6_electrolyte.png`) — top-1 and recovered accuracy on IRSpectra-Bench-Electrolyte by battery-electrolyte chemical class (n=46): sp^{3}-C–F easiest (50%), sulfonyl and nitrile hardest (12%); overall 26%/28%.
- **Fig. S5** (`docs/figures/fig_generator_probe.png`) — trained-generator probe (§5.6; a complement, not part of the training-free protocol): candidate recall and deterministic-HOSE top-1 on the 194-compound benchmark for Claude / + scaffold enumeration / + trained generator. Enumeration's near-degenerate isomers collapse the verifier (28.4→16.0%) while the generator's formula-correct candidates convert (28.4→35.1%).
- **Fig. S6** (`docs/figures/fig_verifier.png`) — learned-verifier probe (§5.4; a complement, not part of the training-free protocol). (A) Conditional-on-recall top-1 (n=65, whole benchmark) for the four verifiers: a GNN trained on the same nmrshiftdb2 data as the HOSE lookup reaches the LLM verifier's level (91% vs 89%) where the lookup (85%) does not move off the solver's own ranking. (B) Why: held-out ^{13}C MAE — the learned model is ~2× sharper (1.70 vs 3.23 ppm).

Data and code availability

Everything below is in the project repository. The archival deposit — a complete frozen snapshot of dataset, benchmark, answer keys, predictions, scripts, figure regeneration and the expert-audit package — will carry DOI [**TODO: 10.5281/zenodo.XXXXXXX — mint on submission**]; GitHub is the development mirror and the Zenodo record the citable version.

component	data	regenerated by
IRexp and its structure-complete split	`data/irexp/`, `data/irexp_resolved/`	`spectro_scraper/` (mining pipeline)
benchmark rounds and the within-compound control	`data/benchmark*/`	`scripts/benchmark_v2.py`
ground-truth integrity audit	`data/benchmark*/clean_qids.json`	`scripts/validate_benchmark.py`
headline accuracy (Table 2)	—	`scripts/score_main.py`
battery-electrolyte subset (§4.5)	`data/benchmark_electrolyte/`	`scripts/build_electrolyte_bench.py`, `scripts/score_electrolyte.py`

component	data	regenerated by
formula-only contamination control (§4.6)	data/modality/	scripts/modality_ablation.py
publication-recency control (§4.6)	data/audit/recency_control.json	scripts/contamination_recency.py
forward-verification, original arm (§5.2)	data/fverify/	scripts/forward_verify.py
…extended to all 194 compounds	data/fverify_main/	scripts/forward_verify_main.py, scripts/forward_verify_all.py
generate-wide arm (§5.3)	data/gw/, data/fverify2/	scripts/score_generate_wide.py, scripts/ladder_significance.py
…its coverage gap, closed	data/fverify_gw/	scripts/forward_verify_gw.py
non-LLM verifier comparison (Table 8)	data/fverify/hose_results.txt, data/fverify/verifier_table_results.txt	scripts/hose_predict.py (incl. coverage), scripts/verifier_table.py, scripts/verifier_leakage.py
negative control and selective prediction (§5.5)	—	scripts/verifier_diagnostics.py
what a miss is, and isomer separability (§4.1, §5.1)	—	scripts/analyze_misses.py, scripts/isomer_separability.py
difficulty-threshold sensitivity (§3)	—	scripts/difficulty_sensitivity.py
licence-pool split (§2.1)	—	scripts/split_license_pools.py
recall headroom and scaffold enumeration	—	scripts/analyze_recall_headroom.py, scripts/enumerate_isomers.py, scripts/closing_the_gap.py
blinded expert-audit package (§7)	data/audit/	scripts/make_audit_sample.py
manuscript integrity gates	—	scripts/check_manuscript.py, scripts/verify_statistics.py

Trained complements, both fenced from the training-free protocol. The §5.6 generator ships as a self-contained bundle at `contrib/generator_probe/` — candidates, the de-leaked split with its InChIKey-14 manifest, the verification scripts (`scripts/closing_the_gap_gen.py`, `scripts/forward_verify_gen.py`, `scripts/verify_leakage_exact40.py`) and the blind forward-prediction outputs behind its verified top-1 (`data/fverify_gen/`), so its scorer runs with no missing predictions. The §5.4 learned verifier ships `scripts/gnn_predict.py` (extract /

train / score / control), the trained model `data/nmrshiftdb/gnn_c13.pt`, per-compound results and both leakage checks (`data/fverify/gnn_results.txt`), and the write-up `docs/VERIFIER_PROBE.md`. Model checkpoints are deposited on Zenodo. Both trained predictors of §5.4 need the nmrshiftdb2 dump, which we cannot redistribute; the README gives the one-line fetch.

Companion technical notes: `docs/BENCHMARK.md`, `docs/FORWARD_VERIFY.md`, `docs/MODALITY_ABLATION.md`, `docs/EXPERT_AUDIT_PROTOCOL.md`, `docs/MODELS.md`.

Two cautions for re-users. The public `irexp_release/train` split does **not** hold out IRSpectra-Bench — it overlaps by 117/200 InChIKey-14 — so de-leak with `contrib/generator_probe/build_exp_manifest.py` before training anything that will be evaluated on the benchmark. And the held-out answer keys `data/audit/key.jsonl` and `data/modality/key.json` are deposited but flagged *withhold from blinded reviewers*.

Licensing and attribution

IRexp is derived from two open-access source pools and redistributed under terms compatible with each (see `data/NOTICE`). **(a) Redistribution:** we release only *extracted numeric data* — IR band lists, $^1H/^{13}C$ shift lists, and resolved structures (SMILES/SELFIES/InChIKey) — plus each record's source DOI/accession, not source full text, figures, or PDFs. **(b) Two separable pools:** 119,345 records derive from the PMC Open-Access Subset (**CC-BY-4.0**) and 1,888 from the Chemotion RADAR4Chem FT-IR deposit (**CC-BY-SA-4.0**). The two are separable losslessly by `source_doi` — Chemotion records carry the `10.22000` prefix — and `scripts/split_license_pools.py` materialises them as two files with each record stamped `license`, so users may take the CC-BY pool alone; any combined or Chemotion-derived release carries CC-BY-SA-4.0 to honour the ShareAlike term. Code is released under the MIT License. **(c) Attribution:** re-users must cite this dataset (Zenodo DOI above) and attribute the original publications via each record's `source_doi`.

Author Contributions

I.Y.: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing — original draft. **R.S.:** methodology, software, formal analysis, investigation, validation (trained-generator and learned-verifier probes, §5.4 and §5.6), writing — review and editing. **R.A.V.-H.:** conceptualization, methodology, supervision, writing — review and editing.

Conflicts of Interest

The authors declare no competing interests.

Use of AI tools

This work studies a large language model, and LLMs were also used as instruments and as writing aids; we state both roles explicitly. **As an object of study:** all reported elucidation, forward-prediction and verification results were produced by Claude models invoked under the protocol of §3 and §8 — these are the measurements the paper reports, not assistance in producing it. **As a writing aid:** the authors used an LLM-based coding assistant for figure generation, analysis scripting, manuscript copy-editing, and for the internal review pass that produced several of the corrections recorded in the repository history. **No text, number, figure or citation in this manuscript was accepted without author verification against the released data and**

code; every quantitative claim regenerates from the scripts in `scripts/`. The authors take full responsibility for the content.

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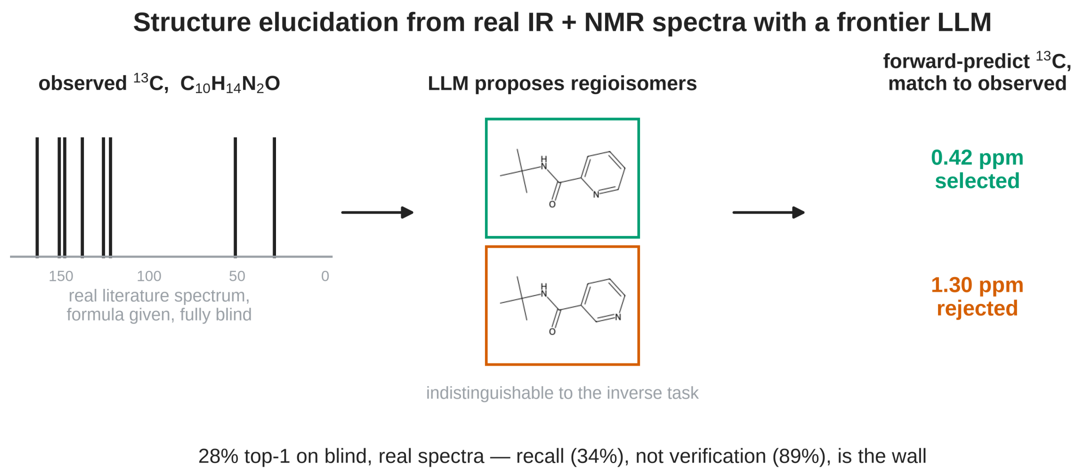
Table of contents entry



A frontier LLM recovers the correct molecular constitution from real, blind IR + ^{1}H + ^{13}C literature spectra for 28% of 194 compounds. The bottleneck is candidate *recall*, not verification: forward-predicting ^{13}C and re-ranking selects the true structure 89% of the time it is proposed — but it is proposed only 34% of the time.